

PRECISION AND RELIABILITY COSTS IN CONTRACTING CONTROL SYSTEMS

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ABSTRACT. We study the number of inputs needed to keep a trajectory near a prescribed set, for two smooth planar maps that both contract toward the origin. The input is chosen once, from a finite catalogue, after the initial state is observed. The allowed distance from a triangle decreases as $\exp(-\gamma n + o(n))$, and the catalogue may fail on initial states of total area $\exp(-\alpha n + o(n))$. For a specified family of first derivatives, we compute the exponential catalogue rate as $\min\{S(\gamma), L_a(\alpha)\}$, provided neither map has a critical point in the triangle. Here S is an explicit precision spectrum and L_a is the classical binary source-coding reliability function. The proof estimates the area served by an arbitrary input and constructs regions on which a given input prefix is forced. Critical sets of area zero suffice for the full-cover spectrum, but not for the reliability formula: an injective C^2 example preserves every finite-horizon full-cover count of the linear model while increasing the reliability lower rate. We also determine the precision rate on a single compact set of arbitrarily nearly full area.

1. INTRODUCTION

A control catalogue is a finite list of input sequences. After observing an initial state, the controller selects one sequence from the list and applies it without further observations. Different states may use the same sequence. The size of the list measures how many choices are needed to maintain a state constraint; its exponential growth is the starting point of invariance entropy [5, 8].

We consider systems in which every input drives the state toward the origin. At first sight this seems to remove the need for a large catalogue. Indeed, at a fixed positive tolerance, all trajectories are eventually close enough to the origin that the constraint becomes automatic. The issue changes when the tolerance decreases with the time horizon. More of the trajectory must then be controlled before contraction makes its remaining motion irrelevant. We ask how many inputs this requires, and how the answer changes if a small set of initial states may be left uncontrolled.

1.1. A linear model. The geometry can be seen in the triangle

$$Q = \{(s, z) : 0 \leq s \leq 1, |z| \leq s\}.$$

For $s > 0$, write $y = z/s$. The first coordinate s parametrises the vertical fibres of the triangle, while $y \in [-1, 1]$ records the slope of the ray through the state. We call y the angular coordinate; it is not an angle in radians. Choose unequal $a_0, a_1 \in (0, 1)$ with $a_0 + a_1 = 1$, and choose $\beta > 1$. The two linear controls act as

control	s after one step	y after one step
0	$a_0^\beta s$	$(y + a_1)/a_0$
1	$a_1^\beta s$	$(y - a_0)/a_1$

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in these coordinates. Thus control 0 is feasible on $[-1, a_0 - a_1]$ and control 1 on $[a_0 - a_1, 1]$. Each expands its own feasible angular interval onto $[-1, 1]$, while shrinking the radius. Every state has a feasible choice at each step.

For a word $w = w_1 \cdots w_m$, put $\lambda_w = \prod_{j=1}^m a_{w_j}$. On any initial fibre $s > 0$, the angles kept in Q by this word form an interval of length $2\lambda_w$, and the final radius is $s\lambda_w^\beta$. These two scales explain the counting problem. Long words serve narrow angular intervals, but their states are also close to the origin. Once the radius is sufficiently small compared with the allowed error, a common continuation can serve them all. A control catalogue can therefore stop distinguishing words before the horizon is reached. The exact stopping construction is given in Section 6.

We use uniform initial-state area on Q , which has total area one. In the linear model the region belonging to w has area λ_w , since the coordinate change has Jacobian s . Consequently words of the same length need not have comparable probabilities. A catalogue allowed to miss rare states can be smaller than one covering the whole triangle. This is where a second budget, for failure probability, enters the problem.

1.2. Precision, reliability and nonlinear controls. An input succeeds through time n if its state stays within Euclidean distance δ_n of Q at every time $0, \dots, n$. It may leave and reenter Q within that tolerance. We take

$$\delta_n \rightarrow 0, \quad -\frac{\log \delta_n}{n} \rightarrow \gamma \geq 0, \quad -\frac{\log e_n}{n} \rightarrow \alpha > 0,$$

where e_n is the permitted initial-state probability of failure. Precision restricts the error along each successful trajectory; reliability restricts the total area of initial states with no successful input in the list.

The nonlinear systems have the first derivatives of the linear model and share its contraction toward the origin. Near the origin, their feasible angular intervals may overlap and have shortened images, and the radius may depend on the angle. If the critical sets have area zero, Theorem 2.1 gives the same full-cover precision spectrum S as in the linear case. This spectrum is linear for small γ , then follows a binary entropy curve, and finally reaches $\log 2$. On one fixed compact set of arbitrarily nearly full area, the rate instead stops at $h = -a_0 \log a_0 - a_1 \log a_1$. The set is chosen once, before the precision exponent or sequence.

If neither control has a critical point on Q , Theorem 2.3 gives the rate

$$\min\{S(\gamma), L_a(\alpha)\}.$$

The two terms have direct constructions: cover every state at the prescribed precision, or keep exactly feasible words while discarding probability at most e_n . The function L_a counts the words needed for the second construction. Both functions are defined and interpreted in Section 2.

The main work is to show that arbitrary approximate inputs cannot produce smaller catalogues. Counting designated feasible words alone would not do this: where feasible branches overlap, choosing a different word need not cause a violation. We estimate a served region by the first time its prescribed input actually exits Q . For the reliability lower bound we also construct disjoint regions on which the first different control already exceeds the tolerance. The regions must have known area, because individual witness points could be discarded at no probability cost.

The distinction between the two critical-point assumptions matters in the quantitative transport of area. A fixed initial part of the trajectory can concentrate probability near rare angular words. Theorem 2.4 constructs such a concentration using an injective C^2 map. It leaves every finite full-cover count unchanged but makes the reliability lower rate strictly larger than the formula above. Critical sets of area zero therefore do not suffice for that formula throughout this class.

The underlying distinction from state chaos is simple: many input choices may be needed even when all resulting feasible states approach one point. The main text quantifies this distinction under uniform exponential contraction by allowing the tolerance to shrink. Appendix D treats the complementary fixed-tolerance question. In that example feasible states converge more slowly, and exact, outer and tracking control entropies are all positive. It belongs to a different contraction regime.

1.3. Relation to earlier work. Our full-cover count is the outer invariance complexity of Chen and Zhong [2, Section 2.2], with a one-step change in the horizon convention. Their Section 3 also treats measure-dependent catalogue complexity. Wang, Huang and Sun [12] define measure invariance entropy using invariant partitions and cylinder probabilities; their Theorem 4.8 transports the initial measure under a control conjugacy. Related probability-dependent formulations include the quasi-stationary entropy of Colonius [3] and the spanning-based entropies and inverse variational principles of Wang and Huang [11].

The probability in the present problem is area on the initial-state triangle. It differs from the measure on the control region obtained by convex duality of invariance pressure in Nie, Wang and Huang [10]. Chen, Huang and Zhong [1] use invariant-partition cylinders and time-power normalisation to study entropy dimension; their high-probability coverings take the failure allowance to zero after the time limit. Here both budgets decrease with the horizon. The estimates below retain that simultaneous dependence, including the probability carried through the first few steps. In particular, transporting a measure under a conjugacy does not compare two systems each equipped with uniform area.

The lower-bound principle of estimating the volume served by one control is already used for invariance pressure in Colonius, Cossich and Santana [4]. Our proofs combine this principle with graph estimates, summable distortion and stopping scales; see also [7, 9]. Once physical coverage and word weights have been compared, the remaining optimisation is the classical method of types [6]. Cost-threshold pressure formulas provide a related counting framework [13]. The specific task here is to control both the error allowed at an actual exit and the area that can be discarded under an exponentially small failure budget.

Section 2 states the results. Sections 3–5 supply the geometric estimates, and Sections 6 and 7 convert them into catalogue rates. The counterexample and further examples occupy Sections 8 and 9. The appendices contain the unequal-power comparison, two finite-horizon comparisons, the slower radial example, and the local derivative calculations.

2. CONTROL CATALOGUES AND MAIN RESULTS

2.1. What a catalogue covers. We keep the triangle

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The list is selected before the initial state; the assigned input may depend on that state. Success means satisfying every inequality in (2.1), including times before a reentry into Q . Only the first n letters matter. Under the controlled-invariance assumption below, the 2^n words already cover Q , so the minimum is finite.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. With the convention $[0, N]$ used in [2, Section 2.2], the exact correspondence is $r_F(n, \delta; K) = r_{\text{inv}}(n+1, \delta, K, Q)$.

2.2. The class of systems. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.3) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

The linear model in the introduction has $\beta_0 = \beta_1 = \beta$. We call this the *matched* case. Allowing the two powers to differ will let us identify its role in Appendix A; the main spectrum and reliability formulas use matched powers. Throughout the paper, except where a separate example is specified, we impose the following assumptions.

- (i) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 maps, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (ii) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (iii) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.4) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

- (iv) Each critical set

$$(2.5) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . The fourth ensures that an exactly feasible finite sequence cannot send a positive-area set to a null set (Lemma 4.2). It still allows folds. In the reliability theorem it will be strengthened to exclude critical points altogether. Conditions on critical points outside Q are unnecessary: the derivative factors used before a first exit are evaluated inside Q .

For comparison, a fixed tolerance has zero exponential cost. Indeed $\|x\|_2 \leq \sqrt{2}V(x)$, and (2.4) gives a fixed J_δ such that every input sends Q into the δ -ball about zero after J_δ steps and keeps it there. Choose an exactly feasible prefix of that length for each initial state and then append an arbitrary tail. The at most 2^{J_δ} prefixes cover Q for every longer horizon. Shrinking precision is what prevents this finite-transient argument from making the asymptotic question trivial.

Both matched and mismatched subfamilies are nonempty. For the linear maps A_i , the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det A_i = r_i^2/a_i > 0$.

2.3. Full coverage and a fixed large initial set. Assume for now that $r_i = a_i^\beta$. The formula for the spectrum uses the frequency p of zero in a word. Define

$$(2.6) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

with $0 \log 0 = 0$. Among length- m words with about pm zeros, the count is $\exp(mH_b(p) + o(m))$, and each has angular weight $\exp(-m\chi(p) + o(m))$. Its radial scale is therefore $\exp(-\beta m\chi(p) + o(m))$. At precision $\exp(-\gamma n)$, one follows such words for a fraction $\min\{1, \gamma/(\beta\chi(p))\}$ of the horizon. This explains the candidate full-cover rate

$$(2.7) \quad S(\gamma) = \max_{p \in [0, 1]} H_b(p) \min\{1, \gamma/[\beta\chi(p)]\}.$$

Theorem 2.1 proves this formula for the nonlinear systems as well. It also compares full coverage with coverage of a compact initial set fixed in advance. Recall $H_b(a_0) = \chi(a_0) = h$.

Theorem 2.1 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.8) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.9) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.10) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.2 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.1, put $a = a_* \in (0, 1/2)$, $c_b = -\log(1 - a)$ and $d_b = \log((1 - a)/a) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.11) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.12) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.13) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.*

The first threshold concerns equality across all positive-area compact initial sets; the second concerns the full-set catalogue. The rate of an arbitrary compact initial set above the first threshold is constrained by (2.9), but is not determined by the theorem.

The role of matching can also be tested within the same matrix family. Appendix A proves that unequal radial powers give distinct full-state and near-full-set rates throughout an interval of positive exponents adjacent to zero. A common positive-area rate on such an interval therefore holds precisely when the powers match. This criterion concerns ordinary precision costs in the specified family; it is not a classification of reliability.

2.4. Allowing an exponentially small failure probability. Let $\mu = \text{area}|_Q$, which is a probability since $\text{area}(Q) = 1$. For $0 < e < 1$, define

$$(2.14) \quad r_e(n, \delta; Q) = \min \left\{ \#\mathcal{U} : \mu \left(\bigcup_{u \in \mathcal{U}} A_u(n, \delta; Q) \right) \geq 1 - e \right\}.$$

The minimum is over finite catalogues of unrestricted binary inputs. It exists because the full-cover count is finite. A failure sequence has $0 < e_n < 1$, $e_n \rightarrow 0$ and $-\log e_n/n \rightarrow \alpha > 0$. Both sequences may be nonmonotone. The covered subset in (2.14) may change with n ; it is not a fixed compact initial set.

For a family of words with zero frequency p , its total reference probability has exponent $\chi(p) - H_b(p)$. Set

$$(2.15) \quad I_a(p) = \chi(p) - H_b(p), \quad L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p).$$

Here $I_a(p) = p \log(p/a_0) + (1-p) \log((1-p)/a_1)$ is the relative entropy of $(p, 1-p)$ with respect to (a_0, a_1) . Frequencies with probability exponent greater than α can be discarded at the exponential scale. Counting the remaining words gives $L_a(\alpha)$, the binary source-coding reliability function. For nonlinear controls, the word weights need not equal initial-state probabilities. The next theorem supplies the required comparison under a stronger derivative assumption.

Theorem 2.3 (Precision–reliability transfer). *Assume the standing hypotheses with $\beta_0 = \beta_1 = \beta$, and strengthen the critical-set condition to*

$$(2.16) \quad \det DF_i(x) \neq 0 \quad (x \in Q, i = 0, 1).$$

For every fixed such system, every finite $\gamma \geq 0$, every $\alpha > 0$, and every precision and failure sequence of those exponents,

$$(2.17) \quad \lim_{n \rightarrow \infty} \frac{\log r_{e_n}(n, \delta_n; Q)}{n} = \min\{S(\gamma), L_a(\alpha)\}.$$

No global injectivity or global inverse is required.

For a finite catalogue, enumerate its inputs and assign a successful state to its first successful input. The sets are Borel, so this is a measurable initial-state encoder. A fixed-length index sent once at time zero has minimum bit length $\lceil \log_2 r_e \rceil$; the asymptotic bit rate is (2.17) divided by $\log 2$. There are no later messages: this is the cost of one initial index. It gives neither a causal feedback data-rate nor a channel capacity.

Theorem 2.4 (A null-critical transient changes reliability). *For $a_0 = 2^{-10}$, $a_1 = 1 - 2^{-10}$ and $\beta = 4096$, there is a fixed injective C^2 pair satisfying the standing matched hypotheses, with critical sets of area zero, such that*

$$(2.18) \quad r_F(n, \delta; Q) = r_A(n, \delta; Q) \quad (n \geq 1, \delta > 0).$$

It maps no positive-area subset of Q into an area-zero set. Nevertheless, for $\gamma = -\beta \log(a_0 a_1)$ and $\alpha = 1/8$, every pair of sequences with these exponents satisfies

$$(2.19) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \frac{1}{2} \log 2 > \min\{S(\gamma), L_a(\alpha)\}.$$

Condition (2.16) is a sufficient condition for (2.17). The counterexample rules out replacing it merely by critical sets of area zero in a universal transfer theorem. It does not show that every system with critical points fails, or give a necessary and sufficient probability criterion.

3. FEASIBLE REGIONS NEAR THE ORIGIN

The linear calculation suggests that a word serves an angular interval of width comparable to its weight. For nonlinear maps we first need to know whether the served set is still an interval. We also need to allow its image to stop short of the full angular range. The estimates below give an upper bound for its width and then a bound for the extra states admitted by a small tolerance.

For a finite word w , write $w|k$ for its first k letters and $F_w = F_{w|w|} \circ \dots \circ F_{w_1}$. Define

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. These are the angular and radial scales. They satisfy $P_w = \lambda_w^\beta$ only in the matched case.

3.1. One step near the origin. Use (s, y) with $z = sy$ for $s > 0$. All local estimates below are on $0 < s \leq \rho$, $|y| \leq 1$, for a small radius ρ chosen below. Write $F_i = (F_i^s, F_i^z)$ and define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

There are constants $B \geq 1$ and $\rho > 0$, depending only on the fixed maps, for which the following estimates hold on $0 \leq s \leq \rho$, $|y| \leq 1$:

$$(3.2) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

We choose ρ small enough to satisfy (E.1) in Appendix E, where the Taylor and quotient calculations giving these constants are recorded. In particular, put $r_* = \min_i r_i$ and $r^* = \max_i r_i$. All constants in this section are fixed before choosing a word. The errors in (3.2) are of order s , and radial contraction makes their sum finite along a feasible trajectory.

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (E.1) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.3) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.3). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.2. Following an interval through several controls. An initially horizontal fibre usually becomes curved after one control, because the new radius may depend on the angle. To keep track of it, we use graphs $s = S(y)$ whose logarithmic slope stays bounded. This bound lets us differentiate the angular map along the moving fibre, rather than at a fixed radius.

Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The image radius is $S(y)R_i(S(y), y)$. Differentiate its logarithm with respect to the new angle, whose derivative is J_i . The result is

$$(3.4) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Thus the new logarithmic slope is again at most one in absolute value: its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (E.1), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. We claim that it is a closed interval, possibly empty or a point. The claim follows by induction, starting with $[-1, 1]$ for the empty word. Suppose the current angular image is an interval whose radial graph satisfies (3.4). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.3), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Let Y_w denote the final angular coordinate as a function of the initial angle on $I_w(s_0)$. Each step changes the linear radial factor and angular derivative by a relative error of order s_j . Since $\sum_j s_j \leq \rho/(1 - \bar{r})$, multiplying over a word produces a bounded distortion, independent of its length:

$$(3.5) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$; the second sums the angular derivatives. To recover the initial interval length, integrate the reciprocal angular derivative over $Y_w(I_w(s_0)) \subset [-1, 1]$. This uses only containment of the image, so it remains valid when the image is shortened. Integrating in the radial variable then gives

$$(3.6) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |I_w| \leq C_A \rho^2 \lambda_w, \quad I_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 9.1 are empty.

Endpoints are included by continuity. A degenerate interval must still be retained in a full cover, although it contributes no area.

3.3. The first true exit. An approximate input can serve more states than its exact feasible region. Estimate these additional states at their first exit from Q . Before that step the trajectory is exactly feasible, so the graph and product estimates apply. At the exit step, success forces the overshoot to be smaller than the prescribed tolerance.

Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least $cs_0 P_w$, by (3.2), (3.5), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.7) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius.

Fix s_0 and regard the current state as a point on the image graph after w . On the side where input i violates the constraint, (3.7) and the radial lower bound give

$$0 < |T_i(S(y), y)| - 1 < \frac{2\delta}{cs_0P_w}.$$

Only one angular boundary can be crossed, by (3.3). The derivative lower bound for T_i along this graph therefore limits this set of current angles to length $C\delta/(s_0P_w)$. Pulling it back by the preceding angular map in (3.5) gives initial-angle length at most

$$C\delta\lambda_w/(s_0P_w).$$

Integration against $s_0 ds_0$ then bounds its physical area by

$$(3.8) \quad C\delta\lambda_w/P_w.$$

The cut point need not lie in the actual terminal image: monotonicity still bounds the part satisfying (3.7), including a point interval. Later reentry is harmless for this estimate, since success requires the inequality at the first exit itself. A change from one feasible coding to another is not an exit.

4. FROM LOCAL GEOMETRY TO COVERED AREA

The estimates of Section 3 apply near the origin, but a catalogue must serve initial states throughout Q . Contraction bounds the number of steps needed to reach that neighbourhood. We call these first steps the *transient*. To pull the local estimates back through it, we need control of inverse area. The null-critical-set assumption does not provide a uniform bound on all of Q . Removing an arbitrarily small fixed area gives a compact set on which every relevant inverse branch has a bounded Jacobian.

The two local area bounds involve λ_w and P_w . When the zero frequency of feasible words tends to a_0 , their logarithmic costs per letter tend to h and

$$(4.1) \quad \chi_R = - \sum_i a_i \log r_i > h,$$

respectively. In the matched case $\chi_R = \beta h$. The following estimate combines the local geometry with the finite initial part of a trajectory. Its three terms correspond to an early exit, exact local survival, and a later first exit.

Theorem 4.1 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ and every $0 < \varepsilon < \text{area}(K)$, one can choose a compact $T \subset K \setminus \{0\}$ with $\text{area}(T) > \text{area}(K) - \varepsilon$ and $C_T < \infty$ so that for every $\eta \in (0, h)$ there is $C_{T,\eta} < \infty$ such that*

$$(4.2) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T\delta + C_{T,\eta} \left[e^{-(h-\eta)m} + \delta e^{(\chi_R - h + 2\eta)m} \right]$$

for every input u , integers $n \geq J$ and $0 \leq m \leq n - J$, and $0 < \delta \leq 1$. The integer depends only on the fixed system. The set and C_T precede η, n, δ, u ; the second constant may also depend on η . In particular, for $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

4.1. Pulling area back through the first few steps. For a finite word w , let

$$(4.3) \quad D_w = \{x \in Q : F_{w|k}x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write w^- for the word with its last letter removed. At a first true exit under w , all derivative inputs still lie in Q , so the relevant domain is D_{w^-} . The last output need not belong to Q .

Lemma 4.2 (Null pullbacks along feasible prefixes). *For every Lebesgue null set $B \subset \mathbb{R}^2$, every control i , and every finite word w ,*

$$(4.4) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

For each nonempty word the compact set

$$(4.5) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

also has area zero.

Proof. We begin with one control. Away from its critical set, it is locally invertible, and each local inverse preserves null sets. More precisely, the inverse function theorem supplies an open inverse chart at every point of $Q \setminus \mathcal{C}_i$. A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its C^1 inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of B is null. The omitted $\mathcal{C}_i$ is null by (2.5). A Borel null superset handles an arbitrary Lebesgue null set.

Now follow a word only while it remains feasible. For $w = vi$, this restriction gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (4.4), including the empty word. Finally, on D_{w^-} the chain rule expresses $\det DF_w$ as a product of determinants evaluated at states in Q . If the factor at time j vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1}\mathcal{C}_{w_j}.$$

This is null by (4.4). Their finite union contains Z_w , which is compact by continuity and closedness of the feasibility domain. $\square$

4.2. One compact set for all precision exponents. The set we retain must work for every horizon and every precision sequence. We obtain it from uniform convergence of word frequencies, before choosing either budget.

Choose $J \geq 1$, depending only on the fixed system, so that

$$(4.6) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible J -prefix ends in $Q_\rho = Q \cap \{s \leq \rho\}$.

Lemma 4.3 (A fixed compact set with uniform estimates). *For every positive-area compact $K \subset Q$ and $0 < \varepsilon < \text{area}(K)$, there is a compact $T \subset K \setminus \{0\}$, with $\text{area}(T) > \text{area}(K) - \varepsilon$, and $L_T < \infty$ such that*

$$(4.7) \quad \text{area}(T \cap D_{w^-} \cap F_w^{-1}B) \leq L_T \text{area}(B)$$

for all $1 \leq |w| \leq J$ and all Lebesgue measurable $B \subset \mathbb{R}^2$. On the same set, for every $\eta \in (0, h)$ there is $D_\eta \geq 1$ such that every local exactly feasible tail v , after any exactly feasible J -prefix from T , satisfies

$$(4.8) \quad \begin{aligned} D_\eta^{-1} e^{-(h+\eta)|v|} &\leq \lambda_v \leq D_\eta e^{-(h-\eta)|v|}, \\ D_\eta^{-1} e^{-(\chi_R+\eta)|v|} &\leq P_v \leq D_\eta e^{-(\chi_R-\eta)|v|}. \end{aligned}$$

The set and L_T precede η , every input, tolerance and horizon.

Proof. Typical frequencies for all feasible words. The weights λ_v are probabilities for independent binary letters with probabilities a_0, a_1 . We first estimate words whose frequencies deviate from a_0 , then use the area bound to transfer that estimate to actual states. For $d > 0$, the total weight $\sum \lambda_v$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$.

Indeed, for a Bernoulli variable X of mean a_0 , the second derivative of $\log \mathbb{E}e^{t(X-a_0)}$ is a tilted variance, at most $1/4$. Its value and derivative at zero vanish, so it is at most $t^2/8$. Exponential Markov with $t = 4d$ and $t = -4d$ gives the bound. By (3.6), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli, applied to rational $d > 0$, gives a Borel null set $B_{\text{loc}} \subset Q_\rho$, including the origin, with the following property. At each point outside it, the maximum deviation $|\#0/m - a_0|$ over all feasible length- m words tends to zero. This is a statement about every feasible word from that point, including possible alternatives in an overlap. The estimate uses a union bound, so those overlaps do not affect it. Controlled invariance ensures that the maximum is over a nonempty family.

Choosing the compact set. We next transfer this almost-everywhere statement through all possible transient words. Lemma 4.2 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1} B_{\text{loc}})$$

is null. The finite union $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$ is a compact null set. On $Q \setminus B_J$, the maximum frequency deviation over all feasible transient choices and all their feasible length- m tails tends to zero. It is measurable: for each m there are finitely many words and their complete feasibility domains are closed. Egorov on K , followed by removal of $B_J \cup Z_J \cup \{0\}$ and compact inner approximation, gives T of the stated area with uniform convergence. Allocate less than $\varepsilon/2$ loss to Egorov and less than $\varepsilon/2$ to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (4.8); finitely many early lengths are absorbed in D_η . This selects one set before all precision choices, rather than a set for each exponent.

A uniform inverse-area constant on that set. For a word w with $1 \leq |w| \leq J$, the compact set $M_w = T \cap D_{w-}$ is disjoint from Z_w . The composition F_w is therefore regular at every point of M_w . Cover it by finitely many open inverse charts $U_{w,k}$, with compact closures inside regular inverse charts. Change of variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1} B) \leq \left(\sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty M_w 's, to obtain (4.7). Chart overlaps only enlarge the bound; every local inverse branch meeting M_w is included. $\square$

The constant L_T may increase when the allowed area loss decreases. For ordinary precision lower bounds this causes no difficulty: T is fixed throughout the horizon limit. The upper bounds will be constructed on all of Q and will include the states removed in selecting T .

4.3. The area served by one approximate input.

Proof of Theorem 4.1. Fix an arbitrary input u . Separate the states it serves according to whether their first exit is before local entry, during the next m steps, or later. We estimate these classes without changing u ; this is what will make the result apply to every possible catalogue. Choose J, T, L_T from Lemma 4.3. Let $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$. For $0 < \delta \leq 1$, the tubes around the three edges of Q give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

Exits during the transient. If a state in $T \cap A_u(n, \delta; K)$ first truly exits at $j \leq J$, put $w = u|j$. Its earlier states lie exactly in Q , and it belongs to

$$T \cap D_{w-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (4.7), its class has area at most $L_T C_Q \delta$. Summing over the J possible first-exit times gives

$$(4.9) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in Q ; no nonsingularity condition is needed at the last output, which may be outside.

States that reach the local region. For the remaining states, $\sigma = u|J$ is exactly feasible. Transport them by F_σ to Q_ρ , and use (4.7) to pull back endpoint area estimates. If the next m prescribed digits are exactly feasible for at least one transported state, (3.6) and (4.8) bound the area of the survival class by $C_{T,\eta} e^{-(h-\eta)m}$. If no such state exists, that class is empty. The states left to count first exit during these m local steps. For such an exit at time $j \leq m$, the prescribed $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (3.8), (4.7) and both products in (4.8), is at most

$$C_{T,\eta} \delta e^{(\chi_R - h + 2\eta)(j-1)}.$$

Since $\chi_R > h$, the sum is bounded by a fixed multiple of $\delta e^{(\chi_R - h + 2\eta)m}$. Together with (4.9) and exact survival, this proves (4.2). The argument only used constraints up to the first exit, so later reentries require no additional estimate. $\square$

5. STATES THAT FORCE A CONTROL PREFIX

To prove a lower bound for full coverage, we shall find many states that require different input prefixes. Two issues arise. A reference angular orbit must be realised by an actual nonlinear trajectory; then the wrong control must move that trajectory farther than the allowed tolerance from Q .

We avoid angular endpoints by using inputs with bounded runs: a run is a consecutive string of equal digits. Bounding its length keeps the reference orbit a definite distance from the endpoints. The bound may be large, but it is fixed while the horizon tends to infinity.

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that, for every $0 < s \leq s_L$, every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at (s, sy_0) . At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The initial-angle centre can be chosen Lipschitz in s , with a constant uniform over the sequences. The radius bound and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. *A reference orbit away from the boundary.* The angular maps expand forward, so it is convenient to construct the orbit using their contracting inverses. These are $f_0(y) = a_0 y - a_1$, $f_1(y) = a_1 y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L+1$ digits. Since $f_0(y) + 1 = a_0(y+1)$, $1 - f_1(y) = a_1(1-y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Solving for the nonlinear orbit. The radius depends on the candidate angular sequence, so an angular inverse iteration alone is insufficient. We first generate the radii from that candidate, then correct the angles using the inverse reference maps. The resulting operator will be a contraction. Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. The bound $|y_k| < 1$ places each candidate in the rectangle where (3.2) holds. Hence these radii stay below $\bar{r}^k s_*$, even before the angular

equation is solved. The scalar map $s \mapsto sR_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.2) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Thus changing the entire candidate angle sequence changes the radii by order s_*^2 . This is small enough to couple the two recursions. Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.2), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its fixed-point equation is $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, the required angular recursion. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

Keeping the other control infeasible. Feasibility of the constructed orbit is not yet enough: a second control might also work. We use the reference margin to rule this out uniformly. Suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.2) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_* d_L/(4a^*)$. Both wrong angular excesses are then at least $d'_L = a_* d_L/(2a^*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system, and the same estimates hold for every $0 < s \leq s_L$. We shall later thicken these orbits over a radial interval, so we also record how the chosen angles vary with the initial radius. For a fixed candidate sequence, scalar contraction gives $\sup_k |s_k(s, y) - s_k(s', y)| \leq |s - s'|$. If $\kappa_L < 1$ is the common contraction constant of (5.3), the fixed-point equations therefore give

$$(5.5) \quad \|y^v(s) - y^v(s')\|_\infty \leq \frac{a^* B}{1 - \kappa_L} |s - s'|.$$

□

An angular excess d'_L at radius s gives a physical excess proportional to s . As the radius shrinks, this distinction eventually becomes smaller than the tolerance. The next lemma counts prefixes only up to a time when every first wrong choice is still detectable.

Lemma 5.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(5.6) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Choose blocks with a prescribed frequency, but fix their first and last digits so that concatenation cannot create long runs. Specifically, use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (3.5), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$. The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(5.7) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (3.7), regardless of later reentry. Thus an input serving a witness must agree with its first m_n digits. The witnesses themselves are distinct: otherwise two selected inputs would both be feasible from one state, contradicting the margin at their first different digit. Each catalogue member therefore serves at most one witness, proving (5.6). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

6. THE FULL-COVER PRECISION SPECTRUM

We now prove Theorem 2.1. For the upper bound, follow an exactly feasible input until either the horizon is reached or the radius is small enough for an arbitrary common continuation. For the lower bound, Section 5 supplies states that force the corresponding number of prefixes. We finish by restricting coverage to a fixed positive-area compact set.

A stopped tree records the finite prefixes used by the construction. Starting from the empty word, continue each branch until its radial product is at most the threshold, or until the permitted depth is reached. The words where the branches stop are its leaves. The tree contains all binary words subject to this rule; some may have no feasible states in the nonlinear system.

Lemma 6.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_R \rho\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.5) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta / M \leq \delta / (4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (2.4) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (6.1). $\square$

6.1. Counting the stopping leaves. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. The counting uses only the length of a leaf and its number of zeros. A class of words with the same frequency is also called a *type*. There are only polynomially many length/type pairs, so their largest exponential count determines the rate. Matching gives $P_w = \lambda_w^\beta$ and $\chi_R = \beta h$. Put

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(6.2) \quad \ell\chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta\chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. Endpoint types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

Only pairs realised by actual initial states are needed for this upper bound.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Forcing enough prefixes. The previous count includes all reference words, whether realised or not. To match it from below, approximate each frequency by the bounded-run blocks of Lemma 5.2. The boundary digits cost only a subexponential factor as the block length grows. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 5.2 is now $c_L = \beta\chi_L$. Fix L first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (6.4) give the zero limit. This proves (2.8).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.1, apply Theorem 4.1 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $\eta \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + \eta} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + \eta)m_n}$. Each term on the right of (4.2) is then at most a constant times $e^{-(h - \eta)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - \eta) \min\{1, \gamma/(\chi_R + \eta)\}.$$

Since $\chi_R = \beta h$ here, letting $\eta \downarrow 0$ gives the lower bound in (2.9); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.8). The evaluation of S below gives equality when $\gamma \leq \beta h$.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N + 1) \exp \left(N \max_{|p - a_0| \leq d_N} H_b(p) \right) = e^{(h + o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. These words are exactly feasible, so their upper rate h is independent of the requested tolerance. Bound (6.5) also gives upper rate γ/β . The lower bound already proved gives (2.10). The construction of T used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.1 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.2. Reparameterise the maximum by the frequency q of the smaller branch $a = a_*$. In this variable $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/a) + (1 - q) \log((1 - q)/(1 - a)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = a$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at a .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (a, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > a$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(a) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $a < 1/2$, $c_b + d_b/2 > c_b + d_b a = h$. The three pieces match at the thresholds, proving (2.11)–(2.12) and the low-side assertion of Theorem 2.1.

Subtracting (2.10) gives (2.13). Strict increase of H_b on $(a, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

7. CATALOGUES WITH A FAILURE ALLOWANCE

Throughout this section $r_i = a_i^\beta$ and neither map has a critical point on Q . We prove Theorem 2.3. Full coverage already costs at most $S(\gamma)$. To use the failure allowance, we must compare word weights with area without first removing a fixed exceptional set: its area would eventually exceed e_n . Nonsingularity provides the required estimate on all feasible transient branches.

For the converse bound, the single witness for each prefix in Section 5 must be replaced by a region of positive area. The total area of those regions determines which families of prefixes the failure allowance can remove.

7.1. A probability bound without an exceptional set. Let J and ρ be as in Section 4, so every exactly feasible J -prefix ends in Q_ρ . Recall D_w , the complete exactly feasible domain, and w^- , the word with its last digit removed.

Lemma 7.1 (Uniform area transport through feasible prefixes). *There is a fixed $K_J < \infty$ such that for every $1 \leq |w| \leq J$ and measurable $B_0 \subset \mathbb{R}^2$,*

$$(7.1) \quad \mu(D_{w^-} \cap F_w^{-1} B_0) \leq K_J \text{area}(B_0).$$

Consequently, for every family $\mathcal{V}$ of words of a common length N ,

$$(7.2) \quad \mu\left(\bigcup_{\substack{|\sigma|=J \\ v \in \mathcal{V}}} D_{\sigma v}\right) \leq C_0 \sum_{v \in \mathcal{V}} \lambda_v, \quad C_0 = \max\{1, 2^J K_J C_A \rho^2\}.$$

The constants are independent of N , the word family, tolerance, failure budget, and horizon.

Proof. Compactness gives $d_0 = \min_{i,x \in Q} |\det DF_i(x)| > 0$. On D_{w^-} all inputs to the derivative factors of F_w belong to Q , even when its last output leaves Q . Thus $|\det DF_w| \geq d_0^{|w|}$ there. Cover this compact domain by finitely many local inverse charts for F_w , with compact closures inside regular charts. Change of variables bounds the pulled-back area in each chart by its inverse-determinant supremum times $\text{area}(B_0)$. Sum over the charts and maximise over the finitely many words of length at most J . All inverse branches meeting the domain are included; their overlap only enlarges the upper bound. No global inverse is used.

An exactly feasible σ sends D_σ into Q_ρ . If σv is feasible, the state after σ lies in the complete initial region for the tail v . Precisely,

$$D_{\sigma v} \subset D_\sigma \cap F_\sigma^{-1}(\mathcal{I}_v \cup \{0\}).$$

By (3.6), this target has area at most $C_A \rho^2 \lambda_v$. Its possible point at zero has null pullback by (7.1). Apply that bound and sum over all at most 2^J transients and all v to obtain (7.2). $\square$

The union estimate is enough even when word regions overlap. We will discard a collection of exactly feasible words and bound the area that loses every such choice. Using approximate words at this stage would introduce an error of order δ_n from early exits. That error need not fit the budget e_n , so exact feasibility is essential to this upper construction.

7.2. Two catalogue upper bounds. To obtain the reliability bound, keep words whose total discarded probability is below e_n . This catalogue is exactly feasible through time n on every state it serves, so its construction has no tolerance error to subtract from the failure budget. Fix $\varepsilon > 0$, $n > J$, and $N = n - J$. Keep every exactly feasible length- n prefix σv , with $|\sigma| = J$, whose tail zero frequency p satisfies $I_a(p) \leq \alpha + \varepsilon$. Empty complete regions may be omitted. Append any infinite input to each kept horizon prefix. Every state not covered has an exactly feasible length- n prefix by controlled invariance, and all such prefixes from that state have discarded tails. By Lemma 7.1, its total probability is at most

$$(7.3) \quad C_0 \sum_{I_a(j/N) > \alpha + \varepsilon} \binom{N}{j} e^{-N\chi(j/N)} \leq C_0(N+1)e^{-N(\alpha+\varepsilon)} < e_n$$

for large n . Here $\binom{N}{j} \leq e^{NH_b(j/N)}$ follows from the j -th term of the binomial probability sum. The catalogue has at most

$$(7.4) \quad 2^J(N+1)e^{NL_a(\alpha+\varepsilon)}$$

members. The fixed transient and polynomial factors have zero rate. Continuity of L_a , proved below, gives upper rate $L_a(\alpha)$.

Independently, the full-state stopping construction in Lemma 6.1 gives upper rate $S(\gamma)$. It assigns an actual feasible prefix to each state; after a threshold leaf its common tail satisfies every remaining constraint. Taking the smaller of these two catalogue bounds yields

$$(7.5) \quad \limsup_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \leq \min\{S(\gamma), L_a(\alpha)\}.$$

7.3. The lower bound below the first precision threshold. For this range, it is enough to retain any fixed positive mass of successful states. The one-input area estimate already bounds the number of controls needed to cover that mass. Take one regular compact $T \subset Q$ of area greater than $3/4$ from Theorem 4.1. For fixed $0 < \varepsilon < h$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta h + \varepsilon} \right\rfloor \right\}.$$

Then $\delta_n \leq e^{-(\beta h + \varepsilon)m_n}$ and, using $\chi_R = \beta h$, (4.2) bounds the area covered inside T by one input by a fixed multiple of $e^{-(h-\varepsilon)m_n}$. A successful catalogue covers more than $1/2$ area inside T eventually. The union bound and $\varepsilon \downarrow 0$ give

$$(7.6) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \min\{h, \gamma/\beta\}.$$

Since $S(\gamma) = \gamma/\beta$ for $\gamma \leq \beta h$ and $L_a(\alpha) \geq h$, this matches (7.5) for $0 \leq \gamma \leq \beta h$. At $\gamma = 0$, the upper bound and $r_{e_n} \geq 1$ give the zero limit. Fixed loss is used here to retain a positive successful mass, not to construct an upper bound with exponentially small failure.

7.4. Positive-area regions forcing complete prefixes. The witnesses in Lemma 5.1 distinguish control words, but individual states have zero area. For a reliability lower bound we need a whole region for each word, with controlled mass and the same forced prefix at every point. The next lemma obtains these regions by thickening the witnesses on the angular scale λ_w .

Lemma 7.2 (Regions with a forced input prefix). *For every integer $L \geq 2$ there are $\tau_L, E_L > 0$ such that any family $\mathcal{W}$ of distinct words of a common length m , each admitting an infinite extension with runs at most L , has pairwise disjoint measurable regions $E_w \subset Q$ with*

$$(7.7) \quad \mu(E_w) = \tau_L \lambda_w.$$

Every point of E_w is exactly feasible under w . If $n \geq m$ and $2\delta < E_L P_w$, every input whose actual successful set $A_u(n, \delta; Q)$ meets E_w has prefix w . All constants are fixed before m, n, δ and the word family.

Proof. The angular expansion through a prefix $w|j$ is at most a constant times $\lambda_{w|j}^{-1}$. An initial interval of width $\epsilon_L \lambda_w$ therefore has width at most a constant times ϵ_L at every time $j \leq m$. We choose ϵ_L to keep this interval inside the feasible region and to preserve the margin for the wrong control. For each word choose a bounded-run extension $v(w)$. Lemma 5.1 gives a common radius bound $s_L > 0$, continuous centres $y_w(s)$ for $0 < s \leq s_L$, and wrong-control excess $d'_L > 0$. Its construction gives centre angles at distance at least $\Delta_L = 3d_L/4$ from both boundaries at every step, where $d_L = 2a_*^{L+1}$. Put $K_0 = 1/a_* + 3B\rho$ and choose

$$0 < \epsilon_L \leq \min\{1/(2C_A), \Delta_L/(4C_A), d'_L/(4K_0C_A)\}.$$

Define

$$(7.8) \quad E_w = \{(s, sy) : s_L/2 \leq s \leq s_L, |y - y_w(s)| \leq \epsilon_L \lambda_w\}.$$

The proposed region is feasible. Fix the initial radius. Initially we may use the derivative estimate only on the feasible component containing the centre. On the interval between the centre and any feasible angle in that component, (3.5) bounds the change in the angle after each prefix $w|j$ by

$$(7.9) \quad |Y_{w|j}(y) - Y_{w|j}(y_w(s))| \leq C_A \epsilon_L \lambda_w / \lambda_{w|j} \leq D_L, \quad D_L = C_A \epsilon_L.$$

To extend this estimate to the whole proposed interval, argue by continuation. If the complete interval $I_w(s)$ ended within the proposed initial interval, continuity would make an initial or intermediate angular constraint attain an endpoint. Its centre is at distance Δ_L , while $D_L \leq \Delta_L/4$, a contradiction. The radii stay positive and below $\rho < 1$, so no radial constraint terminates the interval. Thus the entire interval is feasible. This argument applies even when the complete terminal image is truncated; it does not assume that image is onto.

The other control still fails. At each prefix the thickened state and its centre lie on one image graph with logarithmic radial slope at most one. Hence their log radius ratio has absolute value at most $D_L \leq 1/2$, and their radius difference is at most $2\rho D_L$. The alternative angular map changes by at most

$$(1/a_* + B\rho)D_L + B(2\rho D_L) = K_0 D_L \leq d'_L/4.$$

Every thickened point therefore retains a wrong-control excess at least $d'_L/2$ at every prefix, as well as selected-control feasibility. Continuity of the centres makes E_w measurable.

Area and separation. Integrating with the physical Jacobian s gives

$$2\epsilon_L \lambda_w \int_{s_L/2}^{s_L} s ds = \frac{3\epsilon_L s_L^2}{4} \lambda_w; \quad \tau_L = 3\epsilon_L s_L^2/4.$$

Two distinct equal-length words cannot share a point in their regions: at their first difference one selected control would be wrong, contradicting feasibility under both words.

If an arbitrary input first differs at $j \leq m$, the preceding trajectory is exactly the selected one. Its radius is at least $C_R^{-1}(s_L/2)P_{w|j-1} \geq C_R^{-1}(s_L/2)P_w$. The wrong output has radial factor at least $r_*/2$ and angular excess at least $d'_L/2$. Thus

$$(7.10) \quad |z_j| - s_j \geq E_L P_w, \quad E_L = \frac{s_L r_* d'_L}{8C_R} > 0.$$

Either $z - s$ or $-z - s$ has this positive excess. Each function has Euclidean Lipschitz constant $\sqrt{2}$ and is nonpositive on Q , so the distance is at least $E_L P_w / \sqrt{2} > \delta$. The constraint is violated at that time. Later reentry cannot remove the violation. This proves the all-point forcing assertion. $\square$

7.5. Which rare word families must be covered? Suppose $c = \gamma/\beta > h$. Fix an interior frequency p with

$$(7.11) \quad \chi(p) < c, \quad I_a(p) < \alpha.$$

These conditions express the two obstacles to discarding a family. Its states are still far enough from the origin for different prefixes to be distinguished at precision δ_n , and their total area decays more slowly than the allowed failure probability. We now construct such a family while keeping both inequalities strict. Choose $j_L/L \rightarrow p$ and use the length- L blocks starting with zero, ending with one, and containing j_L zeros. Write

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L), \quad I_L = \chi_L - B_L.$$

The binomial bounds proved in Section 6 give $B_L \rightarrow H_b(p)$, $\chi_L \rightarrow \chi(p)$ and $I_L \rightarrow I_a(p)$. Fix L large enough that $\beta\chi_L < \gamma$ and $I_L < \alpha$. Put $k = \lfloor n/L \rfloor$ and $m = Lk$. The M_L^k concatenations, extended by a fixed allowed block, have runs at most L . Lemma 7.2 gives disjoint regions of equal mass and equal radial product,

$$\mu(E_w) = \tau_L e^{-\chi_L m}, \quad P_w = e^{-\beta\chi_L m}.$$

The physical margin exceeds $2\delta_n$ eventually because its logarithmic ratio has rate $\gamma - \beta\chi_L > 0$. One input can then cover points in at most one region. Hence R inputs cover at most $R\tau_L e^{-\chi_L m}$ of the union, whose total area is $M_L^k \tau_L e^{-\chi_L m}$. At most e_n of this area may be missed, so every successful catalogue satisfies

$$(7.12) \quad R \geq M_L^k - \frac{e_n}{\tau_L e^{-\chi_L m}} = M_L^k \left(1 - \frac{e_n}{\tau_L e^{-I_L m}} \right).$$

The parenthesis tends to one, since $\alpha > I_L$. Take the horizon limit with L fixed, then let $L \rightarrow \infty$. This proves

$$(7.13) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \sup_{\chi(p) < c, I_a(p) < \alpha} H_b(p) \quad (c > h).$$

7.6. Evaluating the probability constraint. It remains to show that the supremum in (7.13) reaches the smaller of the two upper bounds. Relabel by the smaller branch: let $a = a_* < 1/2$ and use q for the frequency of the smaller branch. Define the relabelled functions

$$\hat{\chi}(q) = c_b + dq, \quad \hat{I}(q) = q \log(q/a) + (1-q) \log((1-q)/(1-a)),$$

where $c_b = -\log(1-a)$ and $d = \log((1-a)/a)$, with the endpoint convention $0 \log 0 = 0$. If $a_0 = a$, then $q = p$; otherwise $q = 1-p$. Thus $\hat{\chi}(q) = \chi(p)$, $\hat{I}(q) = I_a(p)$, and $H_b(q) = H_b(p)$. This relabelling changes neither the dynamics nor either variational maximum. Put $J_a = \hat{I}(1/2) > 0$. Then $h = \hat{\chi}(a) = H_b(a)$, and Corollary 2.2 gives the three pieces of S . To evaluate L_a , frequencies $q < a$ have entropy at most h and cannot improve on the feasible frequency a . If $q > 1/2$, reflection preserves entropy and lowers relative entropy:

$$H_b(1-q) = H_b(q), \quad \hat{I}(1-q) - \hat{I}(q) = d(1-2q) < 0.$$

It therefore suffices to optimise over $[a, 1/2]$. On this interval H_b increases, and $\hat{I}$ increases strictly away from its minimum at a . Consequently

$$(7.14) \quad L_a(\alpha) = \begin{cases} H_b(q_\alpha), & 0 < \alpha < J_a, \quad \hat{I}(q_\alpha) = \alpha, \quad a < q_\alpha < 1/2, \\ \log 2, & \alpha \geq J_a. \end{cases}$$

The inverse of this strictly increasing continuous restriction of $\hat{I}$ is continuous; at J_a , q_α tends to $1/2$. Thus L_a is continuous on $(0, \infty)$, including its saturation point. For $c > h$

set $q_c = \min\{1/2, (c - c_b)/d\}$, and set $q_\alpha = 1/2$ if $\alpha \geq J_a$. Every $a < q < \min\{q_c, q_\alpha\}$ satisfies (7.11). Approaching its endpoint in (7.13) yields

$$\liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq H_b(\min\{q_c, q_\alpha\}) = \min\{S(\gamma), L_a(\alpha)\}.$$

Together with (7.5) and (7.6), this proves Theorem 2.3. At the common-cost endpoint $c = h$, the proof uses the fixed-mass argument, while the saturation and reliability endpoints use strict types approaching from below. Integer rounding changes m by less than fixed L ; the transient changes N by fixed J . The strict gaps in (7.3) and (7.12) absorb every $o(n)$ fluctuation in both sequences. No constants are required to be uniform as $L \rightarrow \infty$, or over different fixed systems.

Corollary 7.3 (Linear and fixed-failure specialisations). *The matched linear pair has rate (2.17) for every stated pair of sequences. Under the weaker standing null-critical hypotheses with matched powers, each fixed $e \in (0, 1)$ instead satisfies*

$$(7.15) \quad \lim_n n^{-1} \log r_e(n, \delta_n; Q) = \min\{h, \gamma/\beta\}.$$

Proof. The linear pair is nonsingular on Q , so Theorem 2.3 applies. For fixed failure, choose the single compact set in Theorem 2.1 with area greater than $1 - e$; its catalogue gives the upper rate in (7.15). For the lower rate choose a regular window T of area greater than $(1 + e)/2$. A catalogue with failure at most e covers at least $\text{area}(T) - e > 0$ area there. The fixed-mass proof above, which needs only Theorem 4.1, gives the lower rate. $\square$

8. AN INJECTIVE EXAMPLE WITH A DIFFERENT RELIABILITY COST

We now prove Theorem 2.4. The construction changes area while preserving the correspondence between full covers. Choose a bijection $H : Q \rightarrow Q$ and set $F_i = A_i H$. We arrange that H is the identity near the origin and that every first step from Q reaches this identity region. The trajectory from x then agrees, after time zero, with the linear trajectory from $H(x)$. Full covers have exactly the same size.

To change reliability, H must send relatively large initial regions onto much smaller regions requiring many distinct inputs. We use two Cantor interval constructions with the same binary addresses. Each source interval is much longer than its target. The total source mass will be too large to discard, while the target dynamics force different prefixes on different intervals. The target lengths must also decrease fast enough, relative to the source lengths, for the interval correspondence to extend to a C^2 map. The ratios chosen next meet both requirements.

8.1. Source and target intervals. Fix the parameters of the theorem and put $\lambda = a_0 a_1$. The inverse reference maps are $f_0(y) = a_0 y - a_1$ and $f_1(y) = a_1 y + a_0$. For the control blocks 01 and 10, write

$$B_0 = f_0 \circ f_1 = \lambda y + a_0^2 - a_1, \quad B_1 = f_1 \circ f_0 = \lambda y + a_0 - a_1^2.$$

Their invariant interval has endpoints and length

$$(8.1) \quad A = \frac{a_0^2 - a_1}{1 - \lambda}, \quad B = \frac{a_0 - a_1^2}{1 - \lambda}, \quad \ell = B - A = \frac{2\lambda}{1 - \lambda}.$$

Indeed $A + 1 = 2a_0^2/(1 - \lambda) > 0$ and $1 - B = 2a_1^2/(1 - \lambda) > 0$, so $[A, B] \subset (-1, 1)$. The maps take this interval to its left and right portions of relative length $\lambda < 1/2$. The corresponding Cantor set $\mathcal{C}$ consists of angles with infinite block controls 01, 10.

On $J_\emptyset = [-1/2, 1/2]$ use the source maps

$$b = 2/5, \quad h_0(y) = by - (1 - b)/2, \quad h_1(y) = by + (1 - b)/2.$$

The two contraction ratios serve different purposes. The gap $\lambda < b^2$ will make the interpolation C^2 at Cantor points. At level k , the source intervals retain total length $(2b)^k$; this is the mass scale used in the reliability obstruction. Their Cantor set $\mathcal{D}$ is null, because the total level- k interval length is $(2b)^k \rightarrow 0$. For $\sigma \in \{0, 1\}^k$ put

$$J_\sigma = h_{\sigma_1} \circ \cdots \circ h_{\sigma_k}(J_\emptyset), \quad Z_\sigma = B_{\sigma_1} \circ \cdots \circ B_{\sigma_k}([A, B]).$$

Within each family these closed intervals are separated by positive gaps. Their lengths are b^k and $\ell\lambda^k$, respectively. Both address maps preserve order and are unique, including endpoints. The target Cantor set is null as $(2\lambda)^k \rightarrow 0$.

8.2. Smooth interpolation between the Cantor sets. We map corresponding Cantor points to one another, then fill each gap by a smooth increasing interpolation. The shrinking ratios must control two derivatives at accumulation points of the gaps. Specifically, we construct a strictly increasing C^2 homeomorphism $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$(8.2) \quad g(y) = y \quad (|y| \geq 1), \quad g([-1, 1]) = [-1, 1], \quad g(J_\sigma) = Z_\sigma, \quad g'(y) = 0 \iff y \in \mathcal{D}.$$

Let $E(t) = e^{-1/t}$ for $t > 0$ and zero otherwise, and set

$$\phi(t) = E(t)E(1-t), \quad \Sigma(t) = \frac{\int_0^t \phi(v) dv}{\int_0^1 \phi(v) dv} \quad (0 \leq t \leq 1),$$

extended by zero and one outside that interval. The function is smooth, strictly increasing on $(0, 1)$, and flat at both endpoints. Map each point of $\mathcal{D}$ to the point of $\mathcal{C}$ with the same address. On a source gap (u, v) whose corresponding target gap is (U, V) , define

$$(8.3) \quad g(y) = U + (V - U)\Sigma((y - u)/(v - u)).$$

The ordered endpoint correspondence and increasing interpolation give the full interval identity $g(J_\sigma) = Z_\sigma$. At level k , the gap lengths are $(1 - 2b)b^k$ and $(1 - 2\lambda)\ell\lambda^k$. For $j = 1, 2$ the gap derivatives obey

$$(8.4) \quad |g^{(j)}| \leq C_j(\lambda/b^j)^k.$$

Since $\lambda < 2^{-10} < b^2$, both bounds tend to zero with level. Define q_1, q_2 by these derivatives inside gaps and by zero on $\mathcal{D}$. They are continuous, by (8.4) for high-level gaps and endpoint flatness for the remaining finite collection.

Continuity of the candidate functions q_1, q_2 alone would not identify them as derivatives of g . To do so, recover each function by integration. The integral of q_1 over a gap is its target gap length. Nullity of $\mathcal{C}$ and order preservation therefore imply $g(x) = A + \int_{-1/2}^x q_1(y) dy$ on $J_\emptyset$; inside a gap include the partial integral of (8.3). Moreover

$$\sum_{\text{gaps}} \int |q_2| \leq C \sum_{k \geq 0} (2\lambda/b)^k < \infty.$$

Each complete-gap integral of q_2 is zero, and a partial-gap integral equals $q_1(x)$. Since $\mathcal{D}$ is null, $q_1(x) = \int_{-1/2}^x q_2(y) dy$. The fundamental theorem of calculus gives $g' = q_1$ and $g'' = q_2$. Thus g is C^2 on $J_\emptyset$, flat to second order on $\mathcal{D}$, with positive first derivative on all gaps.

Joining to the identity outside the Cantor construction. On $[-1, -1/2]$ and $[1/2, 1]$, the endpoint values and derivatives are already prescribed. Choose a positive derivative density with the required integral; this preserves monotonicity and gives the correct two derivatives at each join. On $[L, R]$ with prescribed increment $T > 0$, set $d = \min\{(R - L)/4, T/4\}$ and $v(x) = E(x - L)E(R - x)$. For the left join use $c(x) = 1 - \Sigma((x - L)/d)$,

and for the right use $c(x) = \Sigma((x - R + d)/d)$. In either case $\int c \leq d < T$. Integrate the positive interior density

$$(8.5) \quad q(x) = c(x) + \frac{T - \int_L^R c}{\int_L^R v} v(x).$$

For $(-1, -1/2, A + 1)$ start at $g(-1) = -1$; for $(1/2, 1, 1 - B)$ start at $g(1/2) = B$. The identity endpoint has derivatives 1, 0, and the Cantor endpoint has derivatives 0, 0. Extending by identity beyond $[-1, 1]$ therefore gives a global C^2 map satisfying (8.2). It is strictly increasing and onto, and $|g(y) - y| \leq 2$ globally. The map is fixed independently of every horizon and both sequences.

8.3. Planar controls and equality of full-cover counts. The angular map is inserted only at radii above $1/4$. This keeps the first derivatives at the origin unchanged and lets all later steps be linear. Set

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

Define

$$H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + s\psi(s)[g(z/s) - z/s]), & s > 1/4, \end{cases}$$

and set $F_i = A_i \circ H$. The smooth function ψ is zero on $s \leq 1/4$ and one on $s \geq 1/2$. The quotient is used only away from zero; its needed derivatives are locally bounded at every joining point, and flatness of ψ gives global C^2 maps. Near the origin H is identity, so the required first derivatives agree exactly.

Feasibility and contraction. On every fibre the angular map $g_s(y) = (1 - \psi(s))y + \psi(s)g(y)$ is an increasing homeomorphism of $[-1, 1]$ onto itself. Therefore $H(Q) = Q$, and one of the two linear controls is feasible at $H(x)$ for each $x \in Q$. Its radial factor is less than one. This proves controlled invariance including every boundary and the origin.

Put $m_* = \max_i \{a_i^\beta, a_i^{\beta-1}\} < 1$, $\Lambda = 1 + 10m_*/(1 - m_*)$, and $V = \max\{\Lambda|s|, |z|\}$. The bound $|H^z - z| \leq 2|s|$ gives, on the whole plane,

$$V(Hx) \leq (1 + 2/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently the common contraction factor can be taken as

$$\theta = m_*(1 + 1/\Lambda)(1 + 2/\Lambda) \leq m_*[1 + 5/\Lambda] < (1 + m_*)/2 < 1.$$

This applies to arbitrary input tails outside Q as well.

Critical points and injectivity. For $s > 1/4$ the determinant is

$$\det DH = 1 - \psi(s) + \psi(s)g'(z/s), \quad \det DF_i = a_i^{2\beta-1} \det DH;$$

for $s \leq 1/4$ the first determinant equals one. Thus the critical set in Q is exactly $\{(s, sy) : 1/2 \leq s \leq 1, y \in \mathcal{D}\}$. It has area zero by Fubini in physical coordinates. Local inverse charts off that null set show that every null target set has a null pullback in Q . No positive-area set is collapsed to a null set. Each fibre map is strictly increasing and identity outside $[-1, 1]$, so H is globally bijective. It is proper, since its first coordinate is unchanged and $|H^z - z| \leq 2|s|$. Thus it and both F_i are global homeomorphisms, although their Jacobian determinants vanish on the displayed null set.

Why only the first step changes. The fixed radial factors satisfy

$$a_1^\beta = (1 - 2^{-10})^{4096} < e^{-4} < 1/4, \quad a_0^\beta < 1/4.$$

After the first input from Q , the radius lies below $1/4$, regardless of whether its angle is feasible. Every subsequent input therefore uses the identity part of H , even along an approximate trajectory that has left Q . For every input and every $j \geq 1$,

$$(8.6) \quad F_u^j x = A_u^j Hx \quad (x \in Q).$$

At time zero both x and Hx belong to Q . It follows that $A_u^F(n, \delta; Q) = H^{-1}A_u^A(n, \delta; Q)$ on Q . Since $H(Q) = Q$, full covers correspond, proving (2.18). For probability covers the same correspondence transports μ to $H_*\mu$. This measure is absolutely continuous, but it is not uniform area: the interval compression deliberately changes its weights. We next estimate the cost when both systems are instead measured using uniform initial area.

8.4. The mass that a reliable catalogue cannot discard. We first check that each entire target interval forces its block prefix, then compute the area of its source preimage. Every infinite sequence of blocks 01, 10 has runs at most two. The reference endpoint-distance identities used in (5.1) give $|y_j| \leq 1 - d$ at all shifted positions, with $d = 2a_*^3$. For a chosen zero, the current angle is at most $a_0 - a_1 - a_0d$, so the alternative one gives at most $-1 - a_0d/a_1$; the symmetric bound holds for a chosen one. Thus every wrong-control angular excess is at least $d_* = (a_*/a^*)d > 0$. This holds on each complete interval Z_σ through its $2k$ -digit block prefix: its endpoints have such bounded-run extensions, and the linear prefix maps are monotone. No symbolic Cantor subset replaces the whole interval in this assertion.

For each $\sigma \in \{0, 1\}^k$ define an actual physical region

$$E_{\sigma,k} = \{(s, sy) : 1/2 \leq s \leq 1, y \in J_\sigma\}.$$

Here $\psi = 1$ and $g(J_\sigma) = Z_\sigma$. The regions are disjoint and each has mass $(3/8)b^k$, so their union has mass $(3/8)(2b)^k$. Every point requires its selected $2k$ -digit control prefix at sufficiently small tolerance. To check this for an arbitrary input, let its first differing digit be $j \leq 2k$. Equation (8.6) and $s \geq 1/2$ give the preceding physical radius at least $\frac{1}{2}\lambda^{\beta k}$: a partial block product before $2k$ is no smaller than the complete product λ^k . The wrong output radial factor is at least a_*^β and its angular excess is at least d_* . Therefore

$$(8.7) \quad |z_j| - s_j \geq c_0 \lambda^{\beta k}, \quad c_0 = \frac{1}{2} a_*^\beta d_* > 0.$$

Its distance from Q is at least this value divided by $\sqrt{2}$. For $k = \lfloor n/2 \rfloor$ and $\gamma = -\beta \log \lambda$, the logarithmic ratio to δ_n has positive rate $-\frac{1}{2}\beta \log \lambda$. Every successful input thus has the full selected prefix, for every point in the region. Later reentry cannot remove the constraint at this first wrong step.

One catalogue input can consequently cover points of at most one region. A catalogue of R inputs with failure at most e_n obeys

$$(8.8) \quad R \geq 2^k \left(1 - \frac{8e_n}{3(2b)^k} \right).$$

Since $-\frac{1}{2} \log(2b) = \frac{1}{2} \log(5/4) < 1/8 = \alpha$ by $\log(1+x) < x$, the parenthesis tends to one. Thus the lower limit is at least $\frac{1}{2} \log 2$. Both strict exponent gaps absorb all the allowed $o(n)$ fluctuations.

Finally $S(\gamma) = \log 2$: at $p = 1/2$ the threshold ratio $\gamma/[\beta\chi(p)]$ is two. For $q = 1/16$,

$$H_b(q) = \frac{1}{4} \log 2 + \frac{15}{16} \log(16/15) < \frac{1}{4} \log 2 + \frac{1}{16} < \frac{1}{2} \log 2,$$

and

$$I_a(q) > \frac{5}{8} \log 2 - H_b(q) > \frac{3}{8} \log 2 - \frac{1}{16} > \frac{1}{8},$$

using $\log 2 > 1/2$. Relative entropy is strictly increasing for $q > a_0$; its sublevel $I_a(q) \leq 1/8$ therefore has right endpoint below $1/16$. Binary entropy is increasing up to $1/2$, so $L_a(1/8) < H_b(1/16) < \frac{1}{2} \log 2$. This proves the strict inequality (2.19) and completes Theorem 2.4.

The exact reliability spectrum of this example, and whether its rate has a limit, remain undetermined.

8.5. The cost of an exponentially reliable regular window. The regular window in Lemma 4.3 retains a fixed amount of area. One might instead seek windows of mass $1 - e_n$ with inverse-area constants of subexponential size. For this fixed counterexample, the forced regions rule out that approach: their retained mass is much larger than the mass of their first-control images. The following bound holds for every measurable choice of window.

Corollary 8.1. *For the fixed system of Theorem 2.4, let $e_n = \exp(-n/8 + o(n))$, with $0 < e_n < 1$, and $k_n = \lfloor n/2 \rfloor$. For every sequence of measurable windows $T_n \subset Q$ with $\mu(T_n) \geq 1 - e_n$, any finite constants L_n satisfying*

$$(8.9) \quad \mu(T_n \cap F_0^{-1}B) \leq L_n \text{area}(B) \quad \text{for every measurable } B \subset \mathbb{R}^2$$

must, for all sufficiently large n , obey

$$(8.10) \quad L_n \geq \frac{1}{2\ell \det \mathbf{A}_0} (b/\lambda)^{k_n}, \quad \liminf_n \frac{\log L_n}{n} \geq \frac{1}{2} \log(b/\lambda) > 0.$$

Here $b = 2/5$, $\lambda = a_0 a_1$, and $\ell = 2\lambda/(1 - \lambda)$ are unchanged.

Proof. For $k \geq 1$, restrict the regions $E_{\sigma,k}$ above to addresses with $\sigma_1 = 0$, and denote their union by E_k^0 . There are 2^{k-1} disjoint intervals. Since H preserves s and maps each full J_σ onto Z_σ , its image $B_k^0 = H(E_k^0)$ has angular intervals of length $\ell\lambda^k$. The physical Jacobian s gives the exact areas

$$\mu(E_k^0) = 2^{k-1} b^k \int_{1/2}^1 s \, ds = \frac{3}{16} (2b)^k, \quad \text{area}(B_k^0) = \frac{3}{16} \ell (2\lambda)^k.$$

Every target angle in these intervals follows a block word starting with 01 and is exactly feasible under its first zero control. Hence $F_0(E_k^0) = \mathbf{A}_0 B_k^0 \subset Q$; the test uses an actual feasible branch. At $k = k_n$,

$$\frac{1}{n} \log \frac{e_n}{\mu(E_{k_n}^0)} \longrightarrow -\frac{1}{8} + \frac{1}{2} \log(5/4) < 0.$$

Thus every allowed window retains at least half of $E_{k_n}^0$ eventually. Taking the compact measurable set $B = \mathbf{A}_0 B_{k_n}^0$ in (8.9), and using $\det \mathbf{A}_0 = a_0^{2\beta-1} > 0$, gives

$$\frac{1}{2} \mu(E_{k_n}^0) \leq \mu(T_n \cap F_0^{-1}B) \leq L_n \det \mathbf{A}_0 \text{area}(B_{k_n}^0).$$

Division yields the first bound in (8.10). Since $k_n/n \rightarrow 1/2$ and $b > \lambda$, the second follows. The argument includes all the permitted $o(n)$ errors and odd horizons. $\square$

If no finite L_n exists, a subexponential inverse-area bound is already impossible. Thus, for this example and this failure exponent, the fixed-loss estimate of Lemma 4.3 cannot be upgraded by choosing windows with loss e_n and subexponential constants.

9. EXAMPLES AND THE ROLE OF CRITICAL POINTS

The following examples show how to check the hypotheses and why local first derivatives alone do not determine every catalogue cost. The first two have no critical points, so both main formulas apply when the radial powers match. The third has a feasible fold on a null critical set, and the fourth collapses a region of positive area. They distinguish the assumptions used for full coverage, fixed initial sets and reliability.

9.1. Empty fixed-radius slices. In this subsection the powers match: $\beta_0 = \beta_1 = \beta > 1$, so $r_i = a_i^\beta$ and $t_i = a_i^{\beta-1}$. The following example explains why the proof retained complete feasible intervals with possibly truncated images. Its branches overlap, yet long constant words have empty sections at a fixed positive radius. The maps are nonsingular and satisfy the reliability theorem. Their boundary images also rule out a constraint-preserving simultaneous conjugacy to the matched linear pair.

Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1-t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1-v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1+s^2}.$$

Consider the family

$$(9.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their derivatives at the origin are (2.3). For the stated V ,

$$(9.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

This verifies the required common norm inequality on the whole plane.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(9.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

The two interior cut points lie strictly inside $(-1, 1)$, and the overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. Together with nonsingularity and (9.2), this verifies the assumptions of Theorems 2.1 and 2.3.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(9.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. The complete fixed-radius section is therefore empty for sufficiently long all-zero words.

This statement concerns $I_0^m(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_0^m$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Thus every finite word is realised somewhere in Q , despite the empty sections at larger radii.

There is also a constraint-preserving conjugacy obstruction. For (9.1), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(9.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of Q into ∂Q and whose one control sends the whole upper edge into ∂Q , both corresponding intersections contain a continuum. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (9.5) onto the corresponding target intersection, which is impossible because a homeomorphism preserves finiteness. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed Q -preserving simultaneous

control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. This obstruction concerns the stated example and conjugacies that preserve the constraint.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (9.2) gives a global factor at most $67/90 < 1$. Corollary 2.2 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. The spectrum therefore remains valid despite the empty fixed-radius sections.

9.2. Radial motion depending on angle. The graph estimates did not require the radius to evolve independently of the angle. The following nonsingular example makes this dependence explicit while preserving all the standing assumptions. For any parameters (2.2), choose a nonnegative smooth cutoff κ equal to one on $\overline{B}(0, 3/2)$ and supported inside $B(0, 7/4)$. An explicit choice, with the flat function E above, is

$$\kappa(x) = \frac{E(169/64 - |x|^2)}{E(169/64 - |x|^2) + E(|x|^2 - 9/4)}.$$

The denominator is positive, and $\kappa(x) = 0$ for $|x| \geq 13/8$. Put $\phi(s, z) = z^2 \kappa(s, z)$, $M_\phi = \max\{1, \|x \cdot \nabla \phi\|_\infty\}$, choose $0 < c < 1/(2M_\phi)$, and define

$$(9.6) \quad H(x) = e^{-c\phi(x)}x, \quad F_i(x) = A_i H(x).$$

These are global smooth diffeomorphisms. Indeed, on each ray $x = te$, $t \geq 0$, the image radius has derivative $e^{-c\phi(te)}[1 - cte \cdot \nabla \phi(te)] > 0$ and tends to infinity, since H is identity outside a compact set. It maps each ray bijectively onto itself. Moreover

$$\det DH(x) = e^{-2c\phi(x)}[1 - cx \cdot \nabla \phi(x)] > 0.$$

The inverse function theorem gives a smooth inverse, including at zero, and $DH(0) = I$ because $\phi(x) = O(|x|^2)$ there. Thus the first derivatives are (2.3) and there are no critical points. Since $\phi \geq 0$, $V(Hx) \leq V(x)$ globally; the linear norm estimate in Section 2 verifies (2.4). The scalar factor preserves angle and decreases radius. The linear feasible angular domains therefore still cover each fibre of Q , proving controlled invariance at all its states.

The triangle lies in $B(0, 3/2)$, so on Q the first coordinate is $r_i s e^{-cz^2}$. At a fixed positive s it varies with $z = sy$. Radial motion is consequently not autonomous. With matched powers both Theorems 2.1 and 2.3 apply; with unequal powers the statements of Appendix A apply. The example requires no independence between the angular and radial coordinates.

9.3. A fold inside a feasible branch. The ordinary precision theorem allows critical points on a null set. Here this freedom produces genuine collisions within an open feasible branch, even though every input still contracts the norm toward zero.

Proposition 9.1. *For every parameter choice (2.2), matched or mismatched, there are global C^∞ maps satisfying the standing assumptions which agree with A_i near the origin and identify two distinct interior points of Q . For at least one control both points lie in the interior of its feasible domain $Q \cap F_i^{-1}Q$, and their common output lies in $\text{int } Q$. The maps have critical points in Q , but their critical sets there have area zero.*

Proof. We insert a small angular oscillation away from the origin. It keeps angles inside the triangle but reverses the angular derivative on part of a fibre, producing a fold. Fix $c = 1/5$, let $E(v) = e^{-1/v}$ for $v > 0$ and zero otherwise, and put

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

The denominator is positive, $0 \leq \psi \leq 1$, $\psi = 0$ for $s \leq 1/4$, and $\psi = 1$ for $s \geq 1/2$. Define

$$(9.7) \quad H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + cs\psi(s) \sin(2\pi z/s)), & s > 1/4, \end{cases} \quad F_i = A_i H.$$

The quotient is used only away from $s = 0$. On a neighbourhood of each joining point its derivatives are bounded; flatness of ψ at $1/4$ therefore gives global smoothness. On the open set $\{s < 1/4\}$, H is identity, so the maps equal A_i there, including all derivatives at the origin.

On Q the angular action of H is $g_\tau(y) = y + c\tau \sin(2\pi y)$, where $\tau = \psi(s)$. For $0 \leq y \leq 1/2$ it lies in $[0, 1/2 + c]$; for $1/2 \leq y \leq 1$ it lies in $[1/2 - c, 1]$. Oddness handles negative angles, and $g_\tau(\pm 1) = \pm 1$. Thus $H(Q) \subset Q$. At every $H(x)$ one linear control is feasible because its two angular domains cover $[-1, 1]$. The radial factors are less than one, and the origin is fixed. This proves controlled invariance, including all boundary states.

Put $m_* = \max\{r_0, r_1, t_0, t_1\} < 1$ and choose

$$\Lambda = 1 + \frac{2m_*(1 + 2c)}{1 - m_*}.$$

On the whole plane $|H^z - z| \leq c|s|$, and hence

$$V(Hx) \leq (1 + c/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently (2.4) holds globally with

$$\theta = m_*(1 + 1/\Lambda)(1 + c/\Lambda) \leq m_*[1 + (1 + 2c)/\Lambda] < (1 + m_*)/2 < 1.$$

All input trajectories therefore converge to the origin uniformly on bounded initial sets, including trajectories outside Q .

For $s > 1/4$, direct differentiation gives

$$(9.8) \quad \det DH = 1 + 2\pi c\psi(s) \cos(2\pi z/s), \quad \det DF_i = (r_i^2/a_i) \det DH.$$

For $s \leq 1/4$ the first determinant is one. On every fixed positive-radius fibre the determinant has either no zeros or finitely many zeros in $|z| \leq s$. Fubini proves that the critical set in Q has area zero. At $s = 3/4$ the first determinant is positive at $y = 0$ and negative at $y = 1/2$, since $2\pi/5 > 1$. It therefore has actual zeros in Q .

A vanishing determinant by itself does not locate a collision inside a feasible branch. We exhibit one explicitly. At this same radius write $g(y) = y + c \sin(2\pi y)$. Then

$$g(1/4) = 9/20 < 1/2, \quad g(2/5) = 2/5 + (1/5) \sin(\pi/5) > 1/2, \quad g(1/2) = 1/2.$$

The middle inequality uses $\sin(\pi/5) > \sin(\pi/6) = 1/2$. There is $y_1 \in (1/4, 2/5)$ with $g(y_1) = 1/2$. Thus $(3/4, (3/4)y_1)$ and $(3/4, 3/8)$ are distinct interior points with the same image under both controls. Put $\xi = a_0 - a_1$. If $\xi \neq 1/2$, angle $1/2$ is strictly inside one linear feasible branch: use zero when $1/2 < \xi$, and one when $1/2 > \xi$. Its output angle is strictly between -1 and 1 , and its output radius is in $(0, 1)$, so the common output is in $\text{int } Q$.

If $\xi = 1/2$, move the collision off that cut point. Since $g(3/5) = 3/5 - (1/5) \sin(\pi/5) < 1/2$, choose

$$\max\{9/20, g(3/5)\} < v < 1/2.$$

The intermediate value theorem supplies one preimage of v in $(1/4, 2/5)$ and another in $(1/2, 3/5)$. They give distinct interior states at radius $3/4$ with equal images. Now $-1 < v < \xi$, so their common output under zero is in $\text{int } Q$. In both cases continuity and interiority of the initial and output points put each initial point in the interior of $Q \cap F_i^{-1}Q$ for the selected control. $\square$

With matched powers, this fold has the precision spectrum of Theorem 2.1; with unequal powers it satisfies the low-precision conclusions of Appendix A. Its reliability spectrum remains undetermined, since the controls have critical points. The feasible collisions also rule out a Q -preserving simultaneous conjugacy to injective controls: conjugacy preserves injectivity. This conclusion concerns conjugacies, not general semiconjugacies.

9.4. Local agreement with remote positive-area collapse. The next construction agrees with the linear controls on an open neighbourhood of zero and retains all-input convergence, but violates the null-critical-set assumption. It separates local data from the global cost on a fixed positive-area initial set.

Proposition 9.2. *For every matched choice $\beta_0 = \beta_1 = \beta$ in (2.2) and every fixed $0 < \sigma < 1/2$, there are global C^∞ maps G_0, G_1 equal to A_i on $\{s \leq \sigma\}$, with Q controlled invariant and a common global toward-origin norm contraction. The fixed compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and*

$$(9.9) \quad r_G(n, \delta; K_\sigma) = 1 \quad (n \geq 0, \delta > 0).$$

For every finite $\gamma \geq 0$ and every precision sequence of that exponent, the full-state cost satisfies

$$(9.10) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_G(n, \delta_n; Q)}{n} \leq \limsup_n \frac{\log r_G(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular its limit is γ/β for $0 \leq \gamma \leq \beta h$. Its critical sets in Q have positive area.

Proof. We interpolate between the linear controls near zero and maps onto the two boundary rays at larger radii. This leaves a linear core of positive area, while making one input sufficient on the outer part of the triangle. Using the same flat function E , put

$$\psi(s) = \frac{E(s - \sigma)}{E(s - \sigma) + E(2\sigma - s)}.$$

This is smooth, zero for $s \leq \sigma$ and one for $s \geq 2\sigma$. Define, with $r_i = a_i^\beta$ and $t_i = a_i^{\beta-1}$,

$$(9.11) \quad \begin{aligned} G_0(s, z) &= (r_0 s, (1 - \psi(s))t_0(z + a_1 s) - \psi(s)r_0 s), \\ G_1(s, z) &= (r_1 s, (1 - \psi(s))t_1(z - a_0 s) + \psi(s)r_1 s). \end{aligned}$$

There is no division by s . These global smooth maps agree with A_i on the open neighbourhood $\{s < \sigma\}$ of zero. Write $R_0(s, z) = (r_0 s, -r_0 s)$ and $R_1(s, z) = (r_1 s, r_1 s)$. Then $G_i(x) = (1 - \psi(s))A_i x + \psi(s)R_i x$. Choose

$$\Lambda = 1 + \max_i \frac{t_i a_{1-i}}{1 - t_i}, \quad V(s, z) = \max\{\Lambda|s|, |z|\}.$$

The operator bounds are

$$\|A_i\|_V \leq \max\{r_i, t_i + t_i a_{1-i}/\Lambda\} < 1, \quad \|R_i\|_V \leq r_i < 1.$$

Convexity of the norm gives one $\theta < 1$ with $V(G_i x) \leq \theta V(x)$ on all of $\mathbb{R}^2$. Thus every input converges exponentially toward zero uniformly on bounded sets, without an assumption that the trajectory remains in Q .

For $s > 0$ and $y = z/s$, put $t = \psi(s)$ and $\xi = a_0 - a_1$. The output angles are

$$U_0 = (1 - t)(y + a_1)/a_0 - t, \quad U_1 = (1 - t)(y - a_0)/a_1 + t.$$

For $t < 1$ the complete feasible angular domains are

$$J_0(s) = [-1, \min\{1, \xi + 2a_0 t/(1 - t)\}], \quad J_1(s) = [\max\{-1, \xi - 2a_1 t/(1 - t)\}, 1].$$

They follow directly from $-1 \leq U_i \leq 1$ and cover $[-1, 1]$. For $t = 1$ both controls are feasible on the entire fibre. The radius decreases and the origin is fixed, so recursive choice gives an exactly feasible infinite input from every point of Q .

On K_σ a single application of G_0 sends all states to the lower ray. At every radius, $G_0(s, -s) = (r_0s, -r_0s)$, so the same input 0^∞ keeps every point exactly in Q at every subsequent time. The set is nonempty and its area is $\int_{2\sigma}^1 2s ds = 1 - 4\sigma^2$, proving (9.9).

To estimate the full-cover cost from below, retain the inner linear region. Put $c = \sigma/2$ and $Q_c = Q \cap \{s \leq c\}$, whose area is $c^2 > 0$. For every input and every $x \in Q_c$, the radius is $s_k = s \prod_{j=1}^k r_{u_j} \leq c < \sigma$. Therefore $\psi(s_k) = 0$ at all times, independently of the second coordinate. Its entire trajectory is the linear one even if it leaves Q and later reenters. With the same ambient constraint Q this gives

$$(9.12) \quad r_G(n, \delta; Q_c) = r_A(n, \delta; Q_c) \quad (n \geq 0, \delta > 0).$$

The positive-area lower bound of Theorem 2.1, applied to the linear pair and this fixed core, proves the lower bound in (9.10). It concerns arbitrary approximate inputs and every intermediate constraint.

For the upper bound choose a fixed integer $J_\sigma \geq 1$ with $(\max_i r_i)^{J_\sigma} \leq \sigma$. Every state of Q has an exactly feasible prefix of that length, ending in Q with $s \leq \sigma$. Its subsequent dynamics is linear for all later inputs, since the radius continues to decrease even outside Q . Concatenate all such binary prefixes with a linear catalogue on the actual endpoint, using horizon $n - J_\sigma$. The first J_σ times are exactly feasible and the remaining times satisfy the original Euclidean tolerance. By (6.5),

$$r_G(n, \delta; Q) \leq 2^{J_\sigma} r_A(n - J_\sigma, \delta; Q) \leq 2^{J_\sigma} C_U \delta^{-1/\beta} \quad (n \geq J_\sigma, 0 < \delta \leq 1).$$

The common tails in that linear bound satisfy all remaining times; no limit at a shifted precision exponent is assumed. Controlled invariance also gives $r_G(n, \delta; Q) \leq 2^n$. These prove both upper rates and their equality with the lower rate for $0 \leq \gamma \leq \beta h$, including arbitrary $o(n)$ errors. Finally the triangular derivative gives

$$\det DG_i(s, z) = (1 - \psi(s))a_i^{2\beta-1}.$$

It vanishes on K_σ , a positive-area set sent in one feasible step to a null ray. $\square$

For every finite $\gamma > 0$, the full-state lower rate in (9.10) is positive while the same fixed K_σ has zero rate. Decreasing σ makes its area closer to one but changes the system. This is not a single-system claim about zero-cost sets of arbitrarily near-full area. Thus local agreement and contraction under every input do not suffice for the positive-area precision law. The example shows failure after dropping the null-critical-set assumption; it does not establish that assumption as necessary for each individual system. Such collapsed regions are allowed in the general noninjective framework of [2].

10. DISCUSSION

The two budgets constrain different parts of the same counting problem. Precision determines how long distinct input prefixes remain distinguishable in the plane. The failure allowance determines how much initial area associated with those prefixes can be ignored. For $\gamma \leq \beta h$, every positive-area compact set already costs γ/β , so exponentially small failure does not reduce the rate. Above that threshold, a fixed positive failure allowance costs h , while exponential reliability costs $\min\{S(\gamma), L_a(\alpha)\}$ under nonsingularity. It reaches the full-cover cost when $L_a(\alpha) \geq S(\gamma)$.

A finite transient is negligible for word counts but need not be negligible for probability. On a fixed compact set away from the critical pullbacks, its inverse-area constant disappears in the horizon limit. If the exceptional area must decrease as e_n , that constant can instead have a positive exponential rate, as Corollary 8.1 shows. This is the obstruction to deriving the reliability theorem from the weaker full-cover hypotheses.

The formulas concern the stated binary planar family. The general high-precision spectrum for unequal powers and the rate on an arbitrary positive-area compact initial set

above the first threshold remain undetermined. With matched powers the fold has a known full-cover spectrum but no reliability formula here; the collapse example has only bounds above that threshold. The exact reliability spectrum and limit of the Cantor example also remain open. The finite-horizon linear comparison in Appendix C and the slower radial example in Appendix D describe two separate boundaries of the results.

APPENDIX A. MISMATCH AND THE COMMON-COST CRITERION

The main probability results use matched radial powers. This appendix asks whether matching is necessary for a common low-precision rate across all positive-area compact initial sets. When the powers differ, the angular interval size λ_w no longer fixes the radial scale P_w . The earlier geometric estimates kept both products, so they still apply. We now count leaves using their radial weights. For this purpose define

$$\gamma_0 = -\log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

Theorem A.1 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$ and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem A.2 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(A.1) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

Proof of Theorem A.2. Full coverage. For $0 < \gamma < \gamma_0$, even the slowest radial branch reaches the precision scale before the horizon. Thus the depth cap does not affect the stopping count. More precisely, for every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 6.1 cross $P_w \leq \delta_n/M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(A.2) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This gives the upper rate γD with a catalogue satisfying every intermediate constraint. Optimality follows from the lower bound below.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(A.3) \quad H_b(p_*) = D c_R(p_*).$$

Choose $j_L/L \rightarrow p_*$ and use the blocks in Lemma 5.2, with $c_L = c_R(j_L/L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 6 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma/c_L < 1$ therefore gives lower rate $t B_L$. Take its supremum first, then let $L \rightarrow \infty$. Equation (A.3) gives the lower rate

γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (A.1).

A fixed set of positive area. For any fixed positive-area compact K , take the set T in Theorem 4.1. Fix $0 < \eta < h$ and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + \eta} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + \eta)m_n}$. Every term of (4.2) is then bounded by a constant times $e^{-(h - \eta)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - \eta)/(\chi_R + \eta)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $\eta \downarrow 0$. This proves the asserted universal lower bound $\gamma h / \chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 6.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta/M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_\eta}{\chi_R - \eta},$$

by (4.8). At this length, $\lambda_w \geq D_\eta^{-1} e^{-(h + \eta)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_\eta e^{(h + \eta)\ell}$ such words. Sum over these lengths and the at most 2^J transients, and include one input for the origin. For fixed $\eta \in (0, h)$ this gives

$$(A.4) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h + \eta}{\chi_R - \eta} [q_{\delta_n} + c_{\max} + \log D_\eta] + O_\eta(\log(1 + q_{\delta_n})) + O_\eta(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $\eta \downarrow 0$ gives upper rate $\gamma h / \chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before η , the precision exponent, the sequence, and the horizon.

Why the two rates differ. Let $b_i = r_i^D$. The weights b_0, b_1 sum to one, and the relative-entropy identity gives

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h / \chi_R$, completing the theorem. $\square$

Proof of Theorem A.1. If $\beta_0 = \beta_1 = \beta$, Theorem 2.1 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem A.2 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The necessity of matching is established within the first-derivative family (2.3) under the standing assumptions. The argument only uses exponents adjacent to zero; it does not determine the high-precision mismatch spectrum or compare a prescribed strategy with an optimal catalogue.

APPENDIX B. FINITE-HORIZON COUNTS AND THE CONTROL ALPHABET

This finite-horizon example separates equality of exponential rates from equality of catalogue counts when the available controls change. Let $q \geq 2$ be an integer, $0 < \rho < 1$, and consider

$$f_v(s, z) = (\rho s, \rho(qz + sv)), \quad U_q = \{q - 1 - 2j : 0 \leq j < q\}, \quad \hat{U}_q = [-(q - 1), q - 1].$$

Use the same triangle Q , and put $K_1 = [1/2, 3/4] \times [-1/4, 1/4] \subset \text{int } Q$. Here $r_U^{\text{ex}}(n; K)$ denotes the minimum number of input words that keep every assigned state of K in Q

exactly at times $0, \dots, n$, using alphabet U . This illustrative family is separate from the unequal-branch standing assumptions.

For either alphabet,

$$r_U^{\text{ex}}(n; Q) = q^n, \quad q^n/2 \leq r_U^{\text{ex}}(n; K_1) \leq q^n.$$

Indeed, at positive radius the angular map is $y \mapsto qy + v$. For the digit alphabet the inverse one-step intervals tile $[-1, 1]$, and their compositions produce q^n exactly feasible intervals of length $2q^{-n}$, including every intermediate constraint. They give the upper bounds for both alphabets. For any fixed word over the larger alphabet, the final angular map has slope q^n . Its feasible initial angles therefore have length at most $2q^{-n}$. Applying this bound to all of $[-1, 1]$, or to the slice $s = 1/2$ of K_1 with angles $[-1/2, 1/2]$, gives the two lower bounds. Consequently both exact exponential rates are $\log q$ for both alphabets.

The minimum count on K_1 can nevertheless decrease. For $q = 2$ and $n = 1$, input 1 fails on the positive-angle end of K_1 , and input -1 fails on its negative-angle end. Two digit controls are necessary and sufficient. Over $[-1, 1]$, input 0 sends every angle of K_1 to $2y \in [-1, 1]$ and keeps the actual radial coordinate in $(0, 1)$. Thus

$$r_{\{-1, 1\}}^{\text{ex}}(1; K_1) = 2, \quad r_{[-1, 1]}^{\text{ex}}(1; K_1) = 1.$$

Enlarging the control set preserves the displayed full-triangle counts and rate bounds, not all finite-horizon counts.

APPENDIX C. A FINITE-HORIZON PROBABILITY COMPARISON FOR LINEAR CONTROLS

The linear model has an exact interval partition, so its catalogue can be compared with a finite weighted selection problem before passing to exponential rates. Select enough stopping intervals to carry the required probability, and count them. This immediately gives an upper catalogue bound. For the lower bound we show that one arbitrary approximate input can meet only a bounded number of neighbouring intervals. Throughout this appendix the maps are exactly A_i in (2.3), with $r_i = a_i^\beta$. For its inverse angle maps $f_0(y) = a_0y - a_1$ and $f_1(y) = a_1y + a_0$, define for a word $w = i_1 \cdots i_\ell$

$$I_w = f_{i_1} \circ \cdots \circ f_{i_\ell}([-1, 1]) = [d_w - \lambda_w, d_w + \lambda_w].$$

These are its complete feasible initial angle intervals. Each inverse branch maps $[-1, 1]$ into itself, so membership checks every intermediate angle, and all intermediate radii are at most one. For every real initial angle, including infeasible ones, the algebraic trajectory at time ℓ is

$$(C.1) \quad s_\ell = s\lambda_w^\beta, \quad y_\ell = (y - d_w)/\lambda_w.$$

For $n \geq 1$ and $0 < t < 1$, let $\mathcal{L}(n, t)$ consist of the prefixes stopped at the first $\lambda_w \leq t$ or at length n , whichever occurs first. No leaf is a proper prefix of another. The tree has closed leaf intervals with disjoint interiors partitioning $[-1, 1]$. Their weights obey

$$(C.2) \quad \sum_{w \in \mathcal{L}(n, t)} \lambda_w = 1, \quad |I_w| = 2\lambda_w, \quad \lambda_w \geq a_*t.$$

The last bound follows from the parent weight being above t at a threshold leaf; an unfinished horizon leaf has weight above t itself. Shared endpoints and the apex are retained. For $0 < e < 1$ define

$$(C.3) \quad C(n, t; e) = \min \left\{ \#\mathcal{J} : \mathcal{J} \subset \mathcal{L}(n, t), \quad \sum_{w \in \mathcal{J}} \lambda_w \geq 1 - e \right\}.$$

The weights are actual interval masses in the linear model. Indeed, uniform physical probability factors as

$$(C.4) \quad d\mu = s \, ds \, dy = (2s \, ds)(dy/2).$$

The entire physical region over a leaf therefore has mass λ_w .

Lemma C.1. *For each fixed matched linear system there are constants $M > 1$ and an integer $B \geq 1$ such that for every $n \geq 1$, $0 < \delta < 1$, and $0 < e < 3/4$,*

$$(C.5) \quad \frac{1}{B} C(n, \delta^{1/\beta}; 4e/3) \leq r_e^A(n, \delta; Q) \leq C(n, (\delta/M)^{1/\beta}; e).$$

The constants do not depend on the horizon or either budget.

Proof. For the lower bound, we show that one arbitrary input can serve angles in only a bounded number of leaves. Work on the slab $1/2 \leq s \leq 1$, whose probability is $3/4$, and put $t = \delta^{1/\beta}$. For any infinite input u , let v be its stopped prefix in $\mathcal{L}(n, t)$. No feasibility is imposed on this input. The functions $z - s$ and $-z - s$ are nonpositive on Q and Euclidean $\sqrt{2}$ -Lipschitz. At every successful state the constraint at the actual time $|v| \leq n$ therefore implies $(|z_v| - s_v)_+ < \sqrt{2}\delta$. By (C.1), its initial angle satisfies

$$(C.6) \quad \text{dist}(y, I_v) < 2\sqrt{2}\delta\lambda_v^{1-\beta} \leq C_0 t, \quad C_0 = 2\sqrt{2}a_*^{1-\beta}.$$

Here $\beta > 1$ and (C.2) give the last inequality. It remains valid when the input had earlier left Q and reentered; it is a necessary condition at a constrained time.

Expand I_v by $C_0 t$ at both ends. Every other leaf interior has length at least $2a_* t$ and lies outside the interior of I_v . On either side at most $\lceil C_0/(2a_*) \rceil + 2$ leaves can meet the extension: all except a possible first and last one have their interiors entirely in a segment of length $C_0 t$. This also counts closed endpoint contacts. Including v , the expanded interval meets at most

$$B = 5 + 2\lceil C_0/(2a_*) \rceil$$

leaves. This bound uses the extension width $C_0 t$, even when an unfinished horizon leaf itself is much longer than t . For R catalogue inputs let $U \subset [-1, 1]$ be the union of all leaf intervals so encountered. It uses at most BR leaves and contains every successful angle in the slab. Every angle outside U fails on the entire slab, so (C.4) gives $(3/4)\mu_y(U^c) \leq e$, where $d\mu_y = dy/2$. Those leaves have weight at least $1 - 4e/3$, whence $C(n, t; 4e/3) \leq BR$. Minimising over actual catalogues proves the lower comparison for arbitrary approximate inputs.

For the upper bound use the common linear contraction norm from Section 2 and set $M = 4\sqrt{2}\Lambda$. Choose leaves realising (C.3) at $t' = (\delta/M)^{1/\beta}$. Each state over such a leaf has its exact feasible prefix. If the leaf hits the threshold, its terminal norm satisfies

$$V(A_w(s, sy)) = \Lambda s \lambda_w^\beta \leq \Lambda \delta / M = \delta / (4\sqrt{2}).$$

Append the same infinite zero tail. Global toward-origin contraction keeps its Euclidean norm at most $\delta/4$ at every remaining time; since $0 \in Q$, every remaining constraint holds strictly. A leaf stopped only at the horizon is exactly feasible through time n and can have any continuation. The selected physical regions have total probability at least $1 - e$ by (C.4). Endpoints and the apex can use any of their feasible prefixes. One input per selected leaf proves the upper comparison. $\square$

For nonlinear controls the exact interval partition is lost. Section 7 replaces it with a uniform upper bound on word-region probabilities and a lower bound from regions with forced prefixes. Those estimates establish exponential rates, rather than the finite-horizon comparison above.

APPENDIX D. A COMMON RADIAL COMPARISON WITH POSITIVE CONTROL ENTROPY

The fixed-tolerance argument in Section 2 uses convergence to the origin uniformly over all initial states in Q and all inputs. That argument only needs a uniform time after which every trajectory stays inside the tolerance ball; an exponential rate is sufficient but not necessary. Convergence restricted to feasible trajectories does not give this conclusion.

The system below has slower radial decay and positive catalogue entropy at every fixed ambient tolerance, although all its infinitely feasible trajectories approach the origin. We use the exact, outer and reference-trajectory tracking definitions below, with the time limit taken before the tolerance tends to zero.

Let $R(s) = \operatorname{arsinh} s$, $U = \{-1, 1\}$ and

$$(D.1) \quad f_v(s, z) = (R(s), \tau(s)(2z + sv)), \quad \tau(s) = \begin{cases} R(s)/s, & s \neq 0, \\ 1, & s = 0. \end{cases}$$

All infinite inputs $u \in U^{\mathbb{N}}$ are available. Write $\Phi(k, x, u)$ for the actual trajectory, with $\Phi(0, x, u) = x$. For a compact $K \subset Q$, $r_{\text{inv}}(n, K, Q)$ is the minimum size of a finite input family such that

$$\forall x \in K \exists u \forall k \in \{0, \dots, n\} : \Phi(k, x, u) \in Q.$$

The family is selected before x ; only the assigned input depends on x . Replace Q by its Euclidean neighbourhood Q^ε to define $r_{\text{inv}}(n, K, Q^\varepsilon)$. Set

$$h_{\text{inv}}(K, Q) = \limsup_n n^{-1} \log r_{\text{inv}}(n, K, Q),$$

$$h_{\text{out}}(K, Q) = \lim_{\varepsilon \downarrow 0} \limsup_n n^{-1} \log r_{\text{inv}}(n, K, Q^\varepsilon).$$

The time limit precedes the tolerance limit. A tracking catalogue is a finite family (b_j, u^j) whose reference trajectories from $b_j \in Q$ are feasible at every nonnegative time and for which each $x \in K$ has a member satisfying

$$\max_{0 \leq k \leq n} \|\Phi(k, x, u^j) - \Phi(k, b_j, u^j)\|_2 < \varepsilon.$$

Let its minimum size be $r_{\text{inv}}^+(n, \varepsilon, K, Q)$ and define h_{inv}^+ by the same ordered limits as h_{out} . The actual trajectory in a tracking catalogue need not remain in Q ; its reference must. Forgetting references gives $r_{\text{inv}}(n, K, Q^\varepsilon) \leq r_{\text{inv}}^+(n, \varepsilon, K, Q)$. These fixed-tolerance entropy objects are separate from the simultaneous shrinking-precision rates in the main text.

Proposition D.1. *The maps (D.1) are global smooth diffeomorphisms and Q is controlled invariant. For every fixed positive-area compact $K \subset Q$,*

$$(D.2) \quad h_{\text{inv}}(K, Q) = h_{\text{out}}(K, Q) = h_{\text{inv}}^+(K, Q) = \log 2.$$

The exact time limit exists, as do the outer and tracking time limits at every fixed $\varepsilon > 0$. Every infinitely feasible trajectory converges uniformly toward the origin, and two states using the same infinitely feasible input cannot form a state Li–Yorke pair. Here $R'(0) = 1$; this system does not satisfy the uniform exponential norm contraction of the main results.

Proof. Exact feasibility. The smooth positive extension $\tau(s) = \int_0^1 R'(ts) dt$ and the inverse formulas

$$s = \sinh s', \quad z = \frac{1}{2}(z'/\tau(s) - sv)$$

prove global smooth invertibility. For $s > 0$, normalisation $y = z/s$ gives $s_k = R^k(s)$ and $y_{k+1} = 2y_k + u_{k+1}$. The inverse angle branches $(y-1)/2$ and $(y+1)/2$ map $[-1, 1]$ onto its two halves. Iterating them gives 2^n complete feasible intervals of length 2^{1-n} , including every intermediate constraint. They cover all angles, so Q is controlled invariant and all 2^n words cover Q exactly. One word's physical feasible region has area $\int_0^1 s ds 2^{1-n} = 2^{-n}$. Thus

$$\operatorname{area}(K)2^n \leq r_{\text{inv}}(n, K, Q) \leq 2^n, \quad r_{\text{inv}}(n, Q, Q) = 2^n.$$

This proves the exact rate without an interior-margin assumption.

For every fixed $s > 0$, $s_k = R^k(s)$ decreases to zero, since its limit is a fixed point and $0 < R(t) < t$ for $t > 0$. The expansion $R(t) = t - t^3/6 + O(t^5)$ gives

$$s_{k+1}^{-2} - s_k^{-2} \longrightarrow 1/3, \quad R^n(s) \sim \sqrt{3/n}.$$

The second statement follows by telescoping and Cesàro averages.

The lower bound with a fixed tolerance. The decay $R^n(s) \asymp n^{-1/2}$ is too slow to cancel the factor 2^n in the angular coordinate. This gives an area estimate for every approximate input, using just its final-time constraint. Choose $s_0 > 0$ such that $K_0 = K \cap \{s \geq s_0\}$ has positive area. Such a choice exists because $\{s = 0\} \cap Q$ is a null singleton. For any input, feasible or not, the final vertical coordinate as a function of the initial z is affine with slope $2^n R^n(s)/s$. The bounded vertical projection of Q^ε lies in $(-1 - \varepsilon, 1 + \varepsilon)$. Consequently the initial vertical section of states served by one outer input in K_0 has length at most

$$2(1 + \varepsilon)2^{-n} \frac{s}{R^n(s)} \leq \frac{2(1 + \varepsilon)2^{-n}}{R^n(s_0)}.$$

This uses only a necessary final-time condition, hence applies to all actual approximate inputs, including exit and reentry. Fubini and $R^n(s_0) \sim \sqrt{3/n}$ imply that one input covers at most $C_{\varepsilon, s_0} \sqrt{n+1} 2^{-n}$ area in K_0 . Covering its positive area gives lower time rate $\log 2$ at every fixed $\varepsilon > 0$.

Tracking with feasible references. To obtain the matching upper bound, take a finite angle grid in $[-1, 1]$ with mesh less than $\varepsilon 2^{-n}/4$, using at most $C_\varepsilon 2^n$ points. For each centre c choose one infinitely feasible normalised input and denote its angles by $c_k \in [-1, 1]$. Since $0 < R'(s) \leq 1$ on $[0, 1]$, all R^k are 1-Lipschitz there. A finite radial grid, independent of n , therefore ensures $\sup_k |R^k(s) - R^k(t)| < \varepsilon/(4\sqrt{2})$ for an assigned t . Use references (t, tc) with the chosen input of c . Under this common input the actual-reference difference from (s, sy) equals

$$(s_k - t_k, (s_k - t_k)c_k + s_k 2^k(y - c)), \quad s_k = R^k(s), \quad t_k = R^k(t).$$

Its norm is bounded by $\sqrt{2}|s_k - t_k| + s_k 2^k|y - c| < \varepsilon/2$ through time n . Every reference is infinitely feasible, and only a fixed radial factor multiplies the angle-grid count. Thus the tracking upper time rate is at most $\log 2$. The outer lower rate and the count inequality give the matching outer and tracking limits, and then (D.2) in the stated limit order.

Convergence of feasible states. The positive catalogue rates do not prevent all feasible states from approaching one another. For every infinitely feasible input from Q , monotonicity gives $s_k \leq R^k(1)$ and feasibility gives $|z_k| \leq s_k$. All such states lie in $R^k(1)Q$; since $\text{diam } Q = 2$, any two feasible trajectories have distance at most $2R^k(1) \rightarrow 0$. In particular, a common-input state Li–Yorke pair, defined by

$$\liminf_k \|\Phi(k, x, u) - \Phi(k, y, u)\|_2 = 0, \quad \limsup_k \|\Phi(k, x, u) - \Phi(k, y, u)\|_2 > 0,$$

with u infinitely feasible from both states, is impossible. The pair bound also holds with separate feasible inputs, but no convergence of arbitrary infeasible input trajectories is asserted. Finally $Df_v(0) = \begin{pmatrix} 1 & 0 \\ v & 2 \end{pmatrix}$. Its eigenvalues $1, 2$ preclude (2.4) in any norm: that inequality would bound the derivative norm at zero by $\theta < 1$. This comparison is therefore not a special case of the present precision or reliability theorems. $\square$

APPENDIX E. CONSTANTS FOR THE LOCAL DERIVATIVE ESTIMATES

This appendix verifies the choice of B and ρ used in Section 3. It uses only the prescribed derivative at the origin and a local C^2 bound. In particular, none of these constants depends on the word, horizon or precision.

Let L bound the Euclidean bilinear operator norm of both second derivatives on $\overline{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(E.1) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i . In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives (3.2) with the common constant B . For example, $|N_i| \leq 3$ on the stated rectangle, and quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the chosen B bounds every term in (3.2).

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

Generative AI, including ChatGPT (OpenAI), was used extensively in the research and preparation of this draft. Its uses included developing mathematical arguments and examples, checking proof chains and source comparisons, and drafting, restructuring and compiling the English LaTeX manuscript. The research arguments are given in full in the text; AI-assisted checks do not constitute independent verification by a human author. Human authorship, affiliations, the extent of human mathematical review, and approval of the manuscript and this disclosure have not yet been confirmed. These facts must be completed by the responsible human authors before submission.

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